
AKTU B.Tech
SEM-2 : ALL BRANCHES
ENVIRONMENT & ECOLOGY
UNIT-1 IN
ONE SHOT
COMPLETE SYLLABUS
EXAM FOCUSED
EASY TO UNDERSTAND
by MST Sir
GATE Qualified
GW
GATEWAY CLASSES

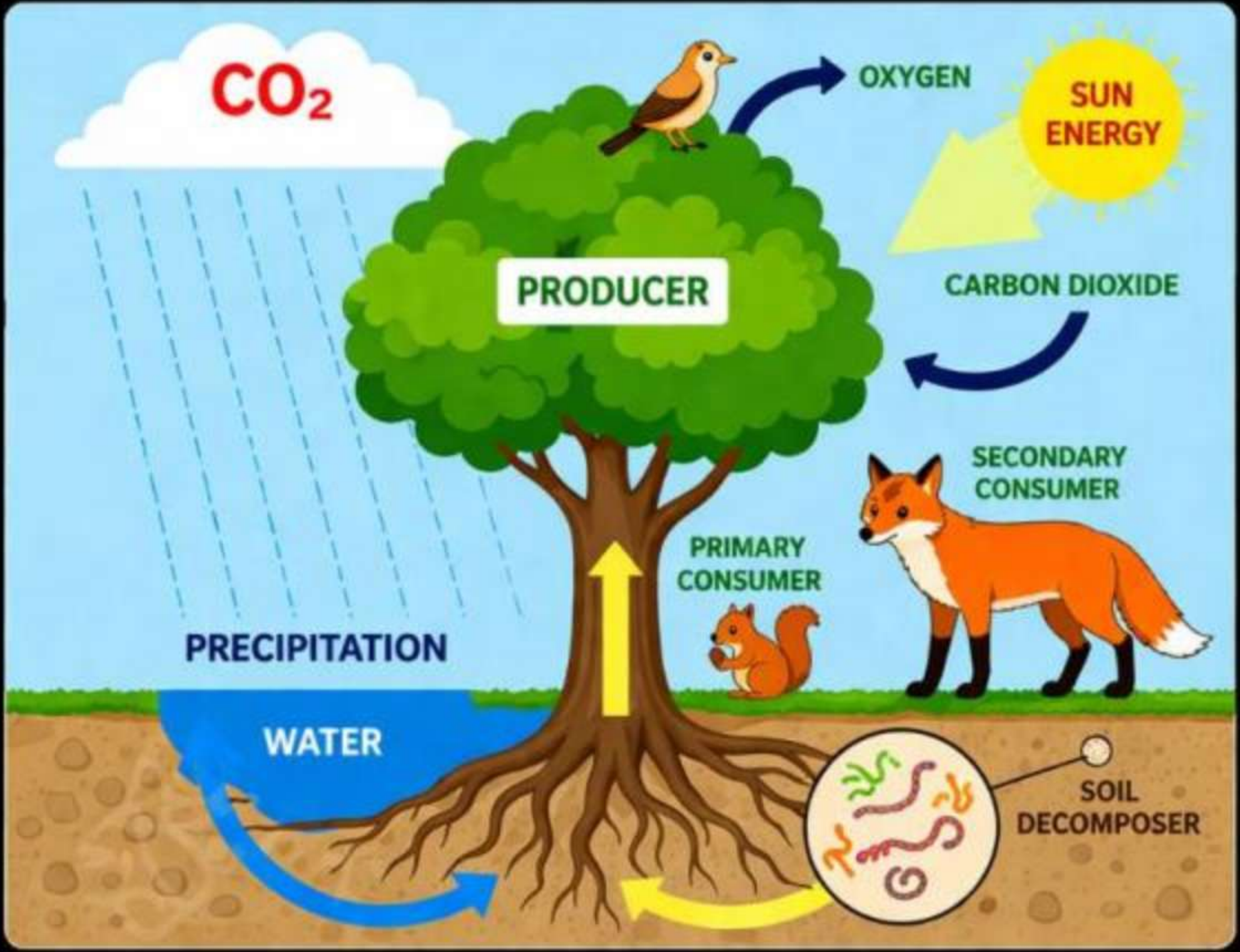
Environment and Ecology

AKTU Syllabus : Unit-1

Environment: Definition, Types of Environment, Components of environment, Segments of environment, Scope and importance, Need for Public Awareness.

Ecosystem: Definition, Types of ecosystem, Structure of ecosystem, Food Chain, Food Web, Ecological pyramid. Balance Ecosystem. Effects of Human Activities such as Food, Shelter, Housing, Agriculture, Industry, Mining, Transportation, Economic and Social security on the Environment, Environmental Impact Assessment, Sustainable Development.

Environment and Ecology



Non-Living (Abiotic)

→ Sunlight
→ CO_2
→ Water
→ Soil

Living (Biotic)

→ Fox
→ Bird
→ Squirrel
→ Plants

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Environment ("पर्यावरण")

⬇

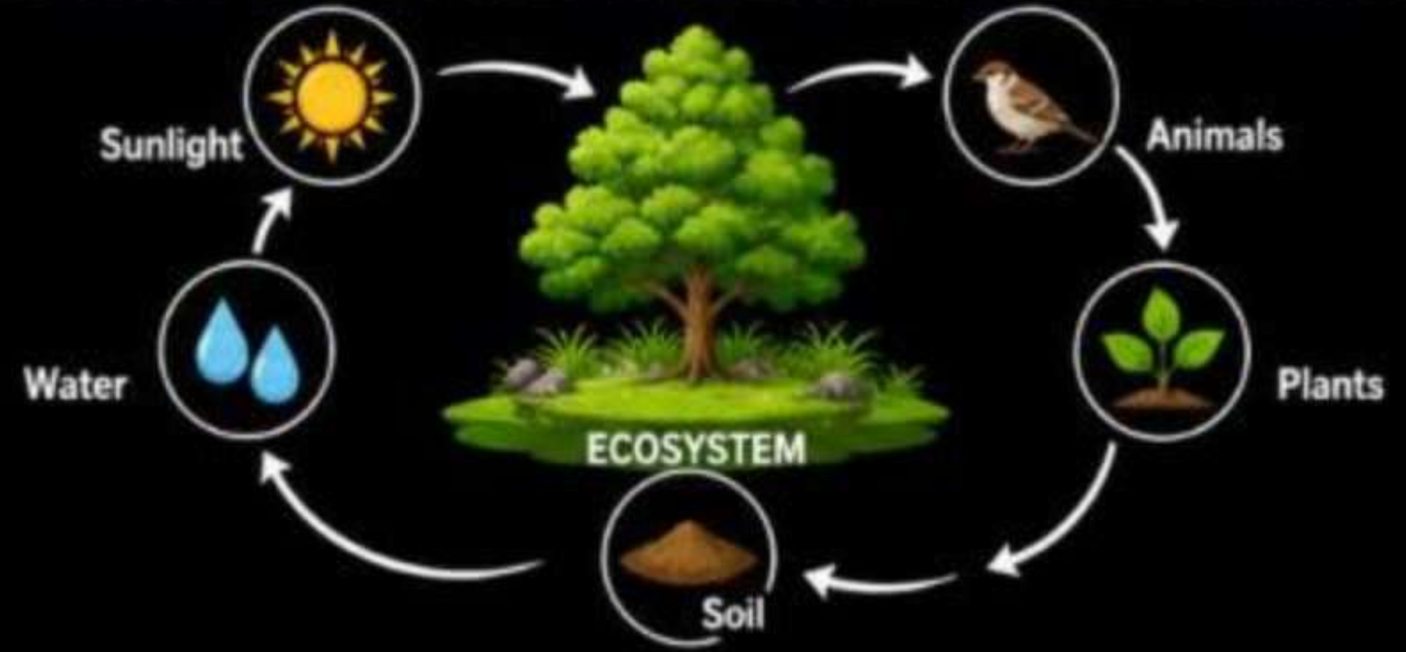
"Whatever we see in our surrounding is called environment"

- **Environment** is the surrounding conditions that affect the growth, development, and survival of living organisms.
- It includes both the natural and man-made surroundings, including air, water, food, sunlight, land, plants, animals, and human beings, road buildings, bridges, etc.
- The environment is our basic life support system.

Q. Who is the father of ecology in India? **AKTU-2034-24 Odd Sem**
Ans. Ramdeo Misra

Ecology ("पारिस्थितिकी")

- Ecology is the study of how living things interact with each other and with their environment.
- It is the relationships between plants, animals, and their surroundings.



Types of Environment

1. Natural Environment ("प्राकृतिक पर्यावरण")

"The natural environment is the surroundings or conditions created by nature without significant human intervention."

- → Air
- → Water
- → Soil
- → Land
- → Forest
- → Wild life

2. Man-made Environment (or Built Environment) ("कृत्रिम पर्यावरण")

"The man-made environment is the surroundings or conditions that have been created by human activity."

- → Housing
- → **Energy Sources:** Hydro, Thermal, Nuclear, Atomic
- → Transportation
- → Agriculture
- → Technology

Components of Environment (पर्यावरण के घटक)

Natural Environment

Biotic (Living) (जीवित)

The world of living organisms

e.g.- Plants and Animals

Abiotic (Non-living) (निर्जीव)

The world of non-living elements.

e.g.- air, water, Land

Human Environment

Activities, Creations and interaction among human beings.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Segments of Environment (पर्यावरण के खंड)

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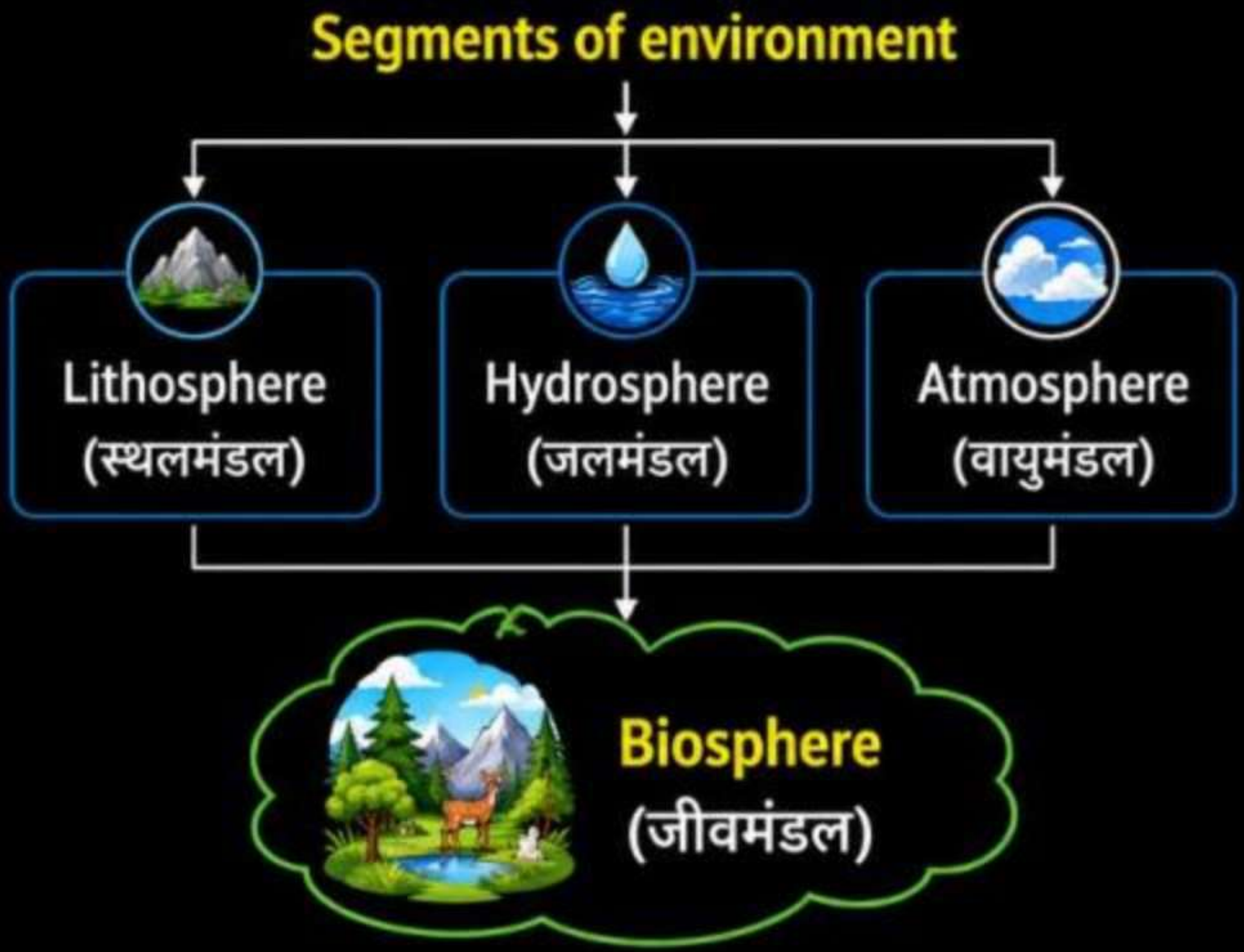
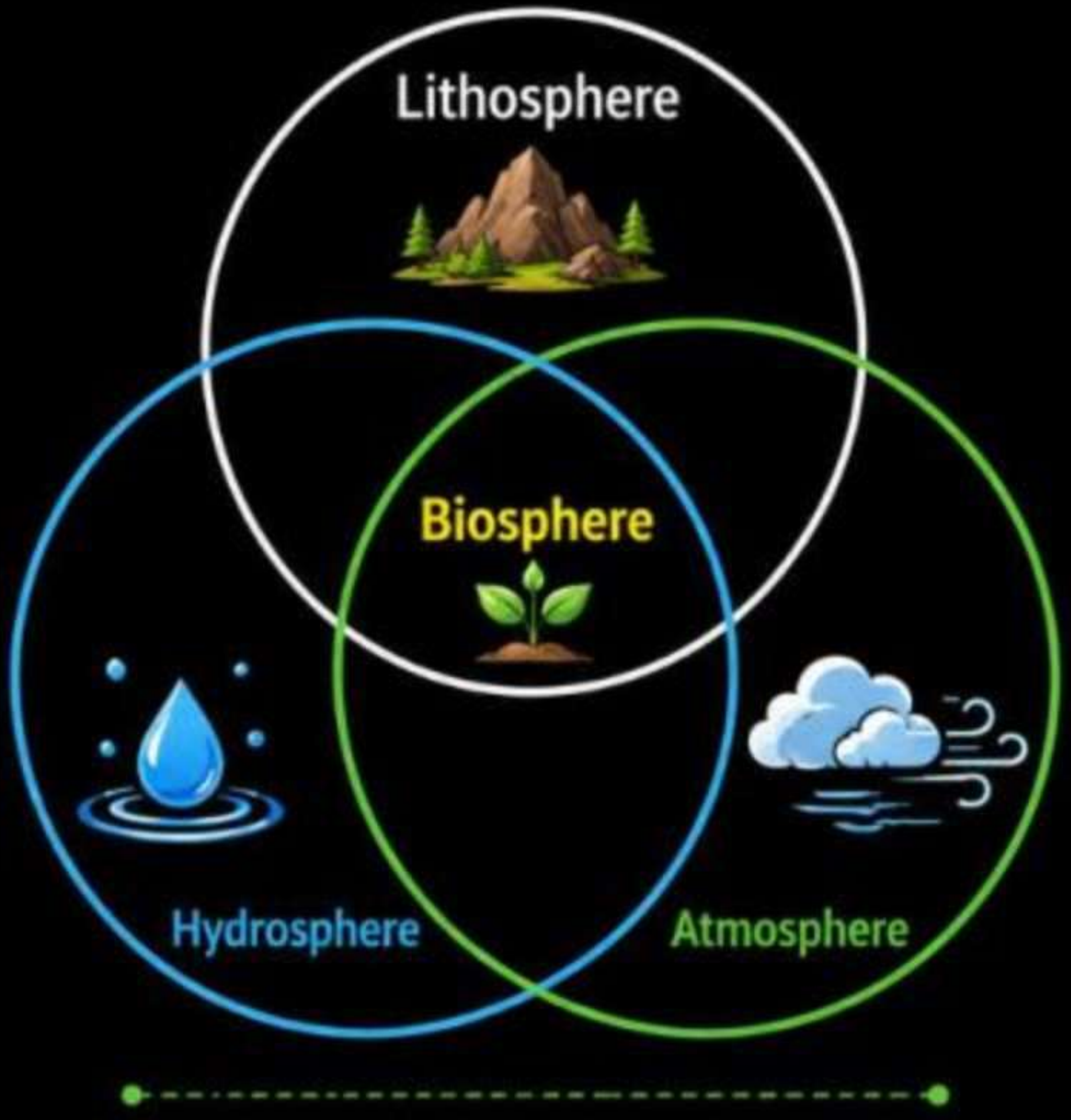
The Environment consists of various segments such as

→ Lithosphere, Hydrosphere, Atmosphere and Biosphere.





Domains of the Environment

...Segments of environment

Lithosphere (स्थलमंडल)

→ Solid crust on the hard top layer of the earth.

→ Made up of rocks and minerals and covered by thin layer of soil.

→ Lithosphere consist of landforms such as mountains, plateaus, plains, valleys, etc.

→ Lithosphere is a domain that provides us forests, grasslands of grazing, land for agricultures and human settlements. we also get minerals from lithosphere.

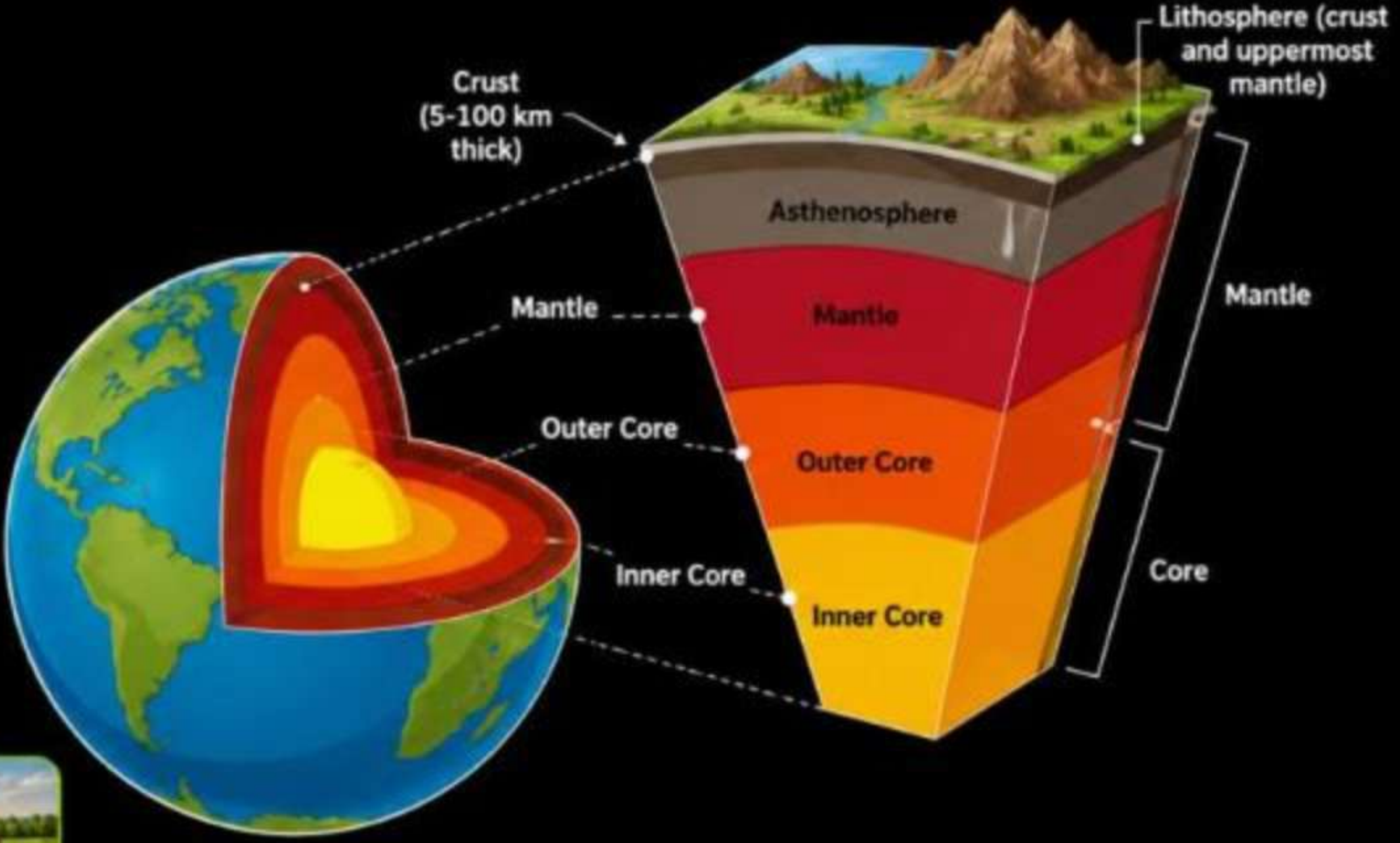

Mountains


Plateaus


Plains


Valleys


Agriculture



Segments of environment

Hydrosphere (जलमंडल)

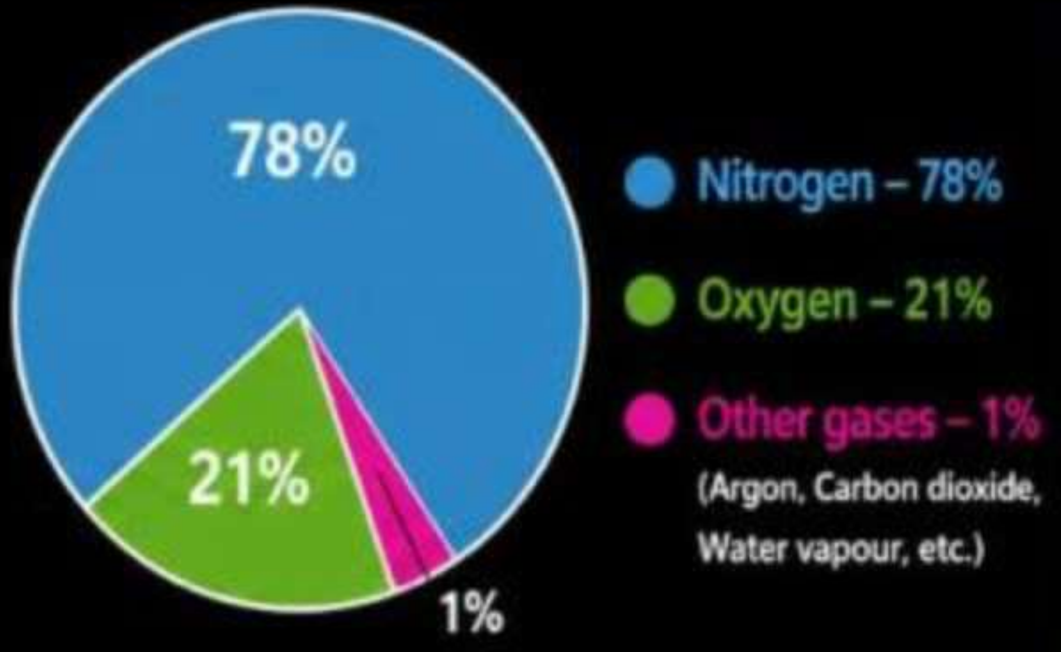
- The domain of water = Hydrosphere.
- It comprises various sources of water and different types of water bodies. e.g. – Rivers, lakes, seas, oceans etc.
- It is essential for all living organisms.

Oceans

Lakes

Atmosphere (वायुमंडल)

- The atmosphere is the layer of gases surrounding earth that supports life.
- The gravitational force of the earth holds the atmosphere around it.
- It protects us from the harmful rays and heat of the sun.
- It consists of number of gases, dust and water vapour.
- Change in atmosphere = change in weather and climate.

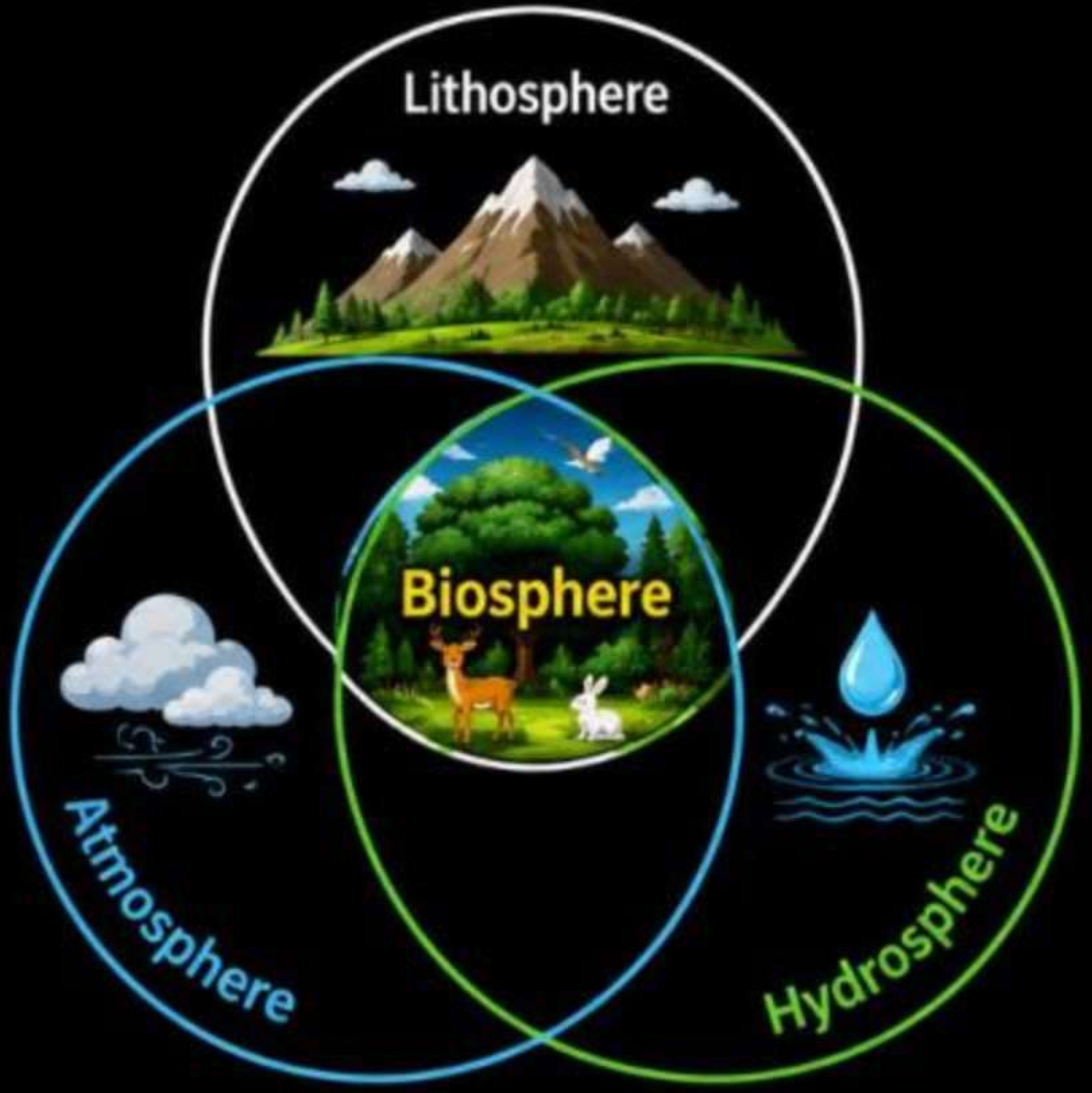


Biosphere (जीवमंडल)

- The biosphere is the part of the earth in which **life exists**.

- It is dependent on the **atmosphere** (air), **hydrosphere** (water) and **lithosphere** (land).

- This denotes the relating of living organism and their **interactions with the environment**.



Classification of Atmosphere
(वायुमंडल का वर्गीकरण)

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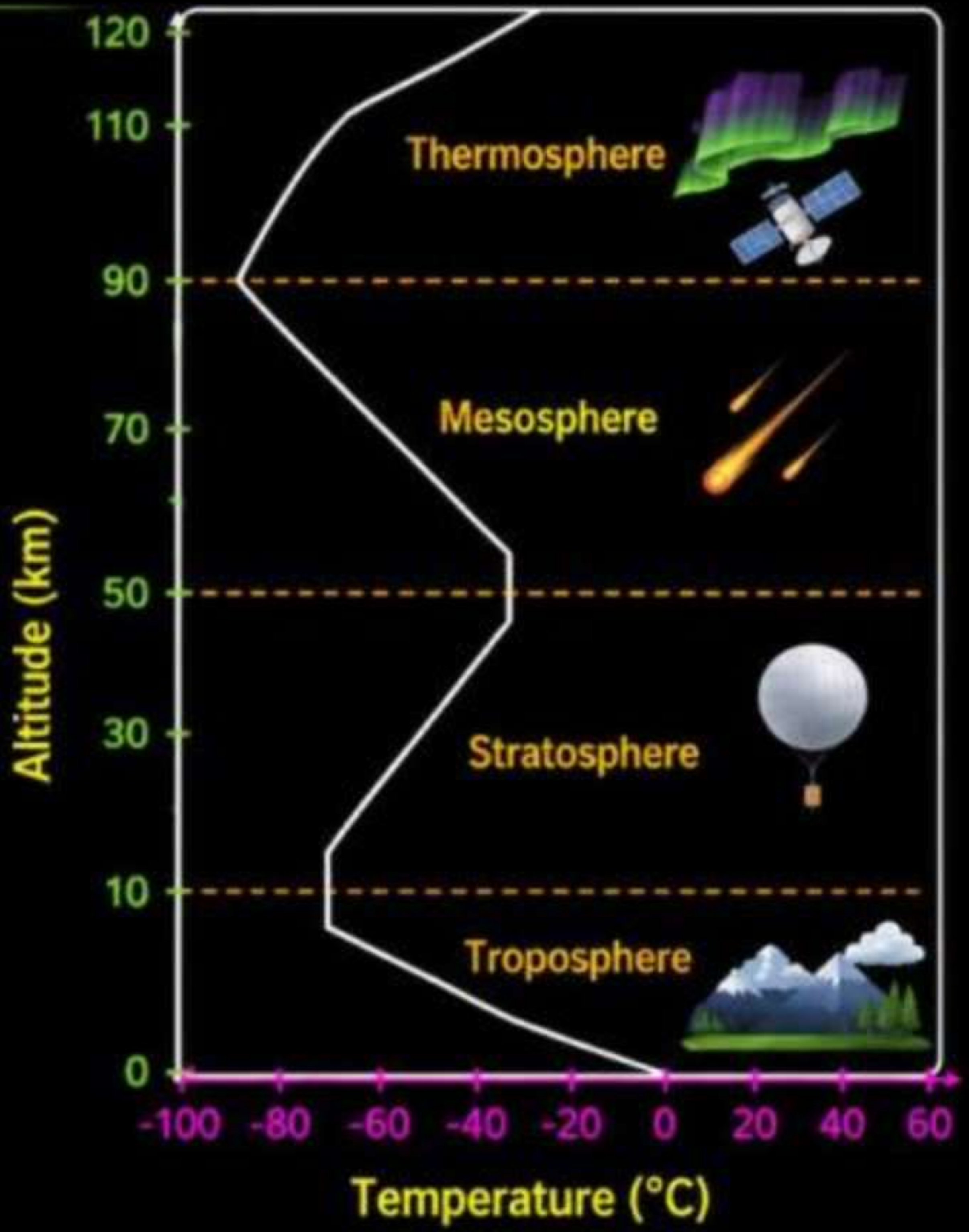
- On the basis of temperature variation, atmosphere is divided into four major layers – **Troposphere**, **Stratosphere**, **Mesosphere** and **Thermosphere**.

1. Troposphere (क्षोभमंडल)

- It is the lower portion of the atmosphere, extending upto about **8 km** at the poles and **16 km** at the equator.

- This is the layer in which most living things exist.

- There is a uniform decrease in temperature with increase in altitude (about **6°C/Km**) to a minimum of – –50°C or - 60°C.



Classification of Atmosphere

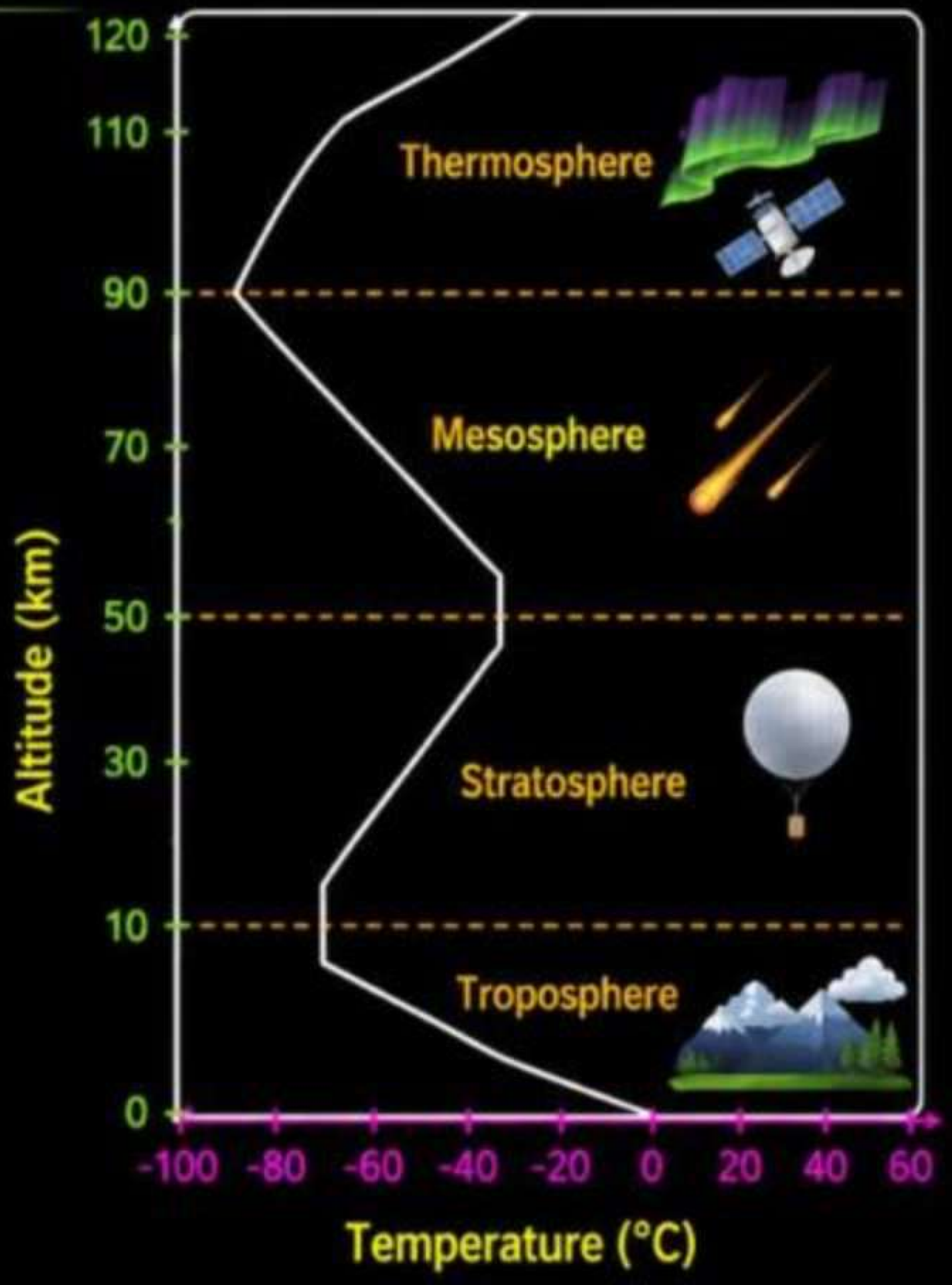
AKTU 2014-15, 2022-23

2. Stratosphere (समतापमंडल)

- It is bove the troposphere; where the temperature is nearly constant upward to about 20 km and then increases upto a maximum of 0°C near its outer limit.
- It contains Ozone which acts as protective layer for the living organisms because it absorbs harmful Ultra Violet (UV) radiations coming from the sun.
- Therefore it is also known as **Ozonosphere**.

3. Mesosphere (मध्यमंडल)

- The Mesosphere lies between the thermosphere and the stratosphere.
- "Meso" means middle, and this is the highest layer of the atmosphere in which the gases are all mixed up rather than being layered by their mass.
- In this layer, the temperature decreases slowly with the altitude but then sharply to a minimum of about -75°C at 80 km.
- They experience increasing friction.



Classification of Atmosphere

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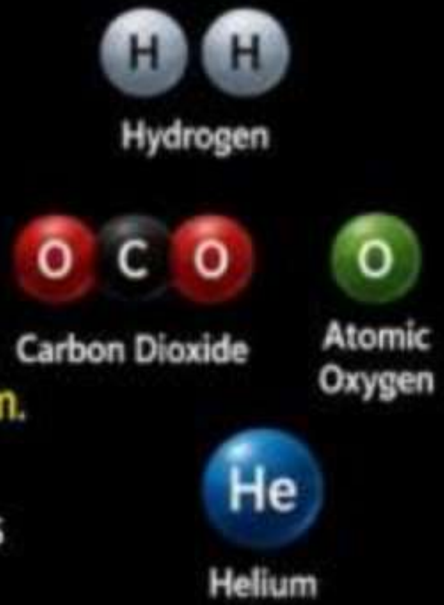
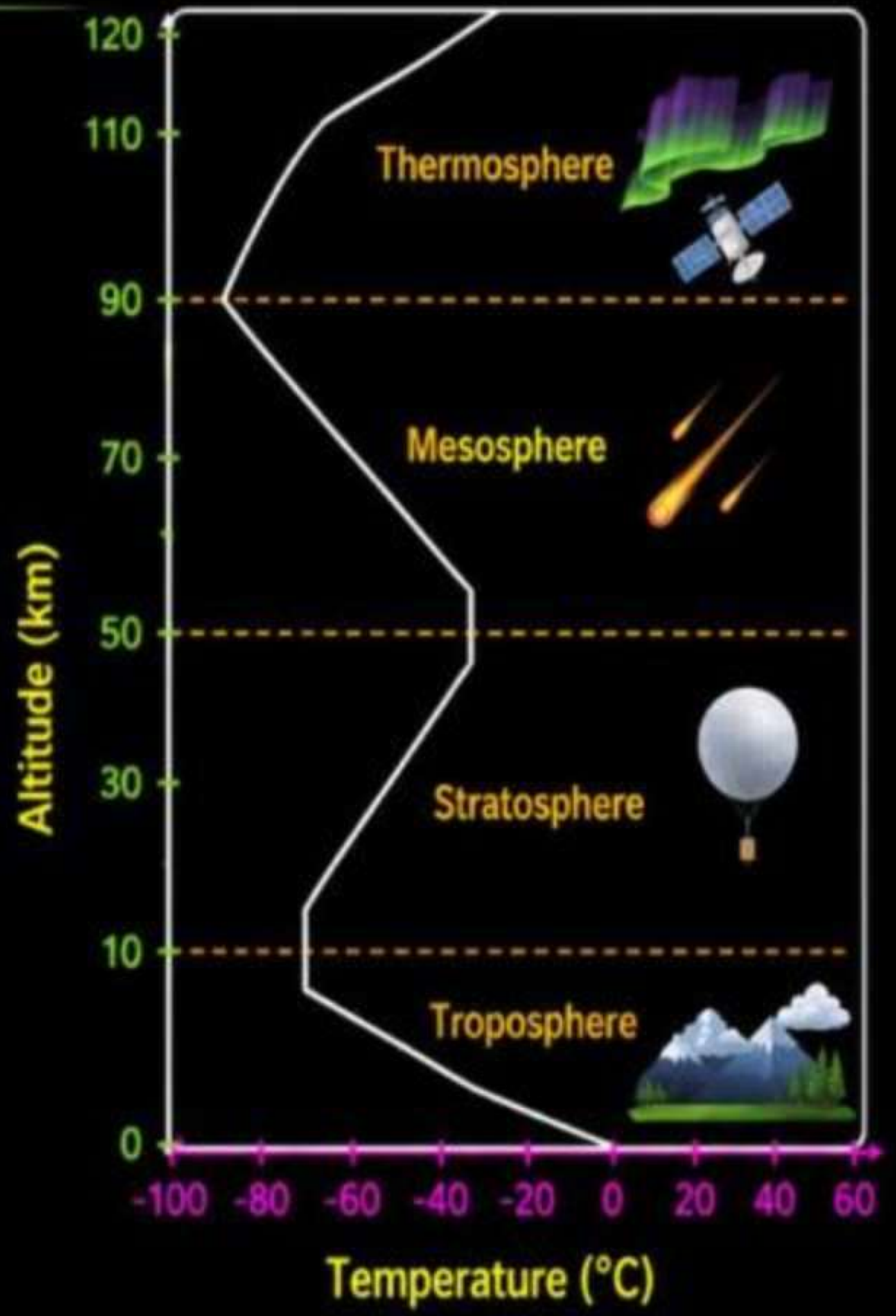
4. Thermosphere (तापमंडल)

- The thermosphere lies between the exosphere and the mesosphere.
- "Thermo" means heat, and the temperature in this layer again rises to a very high value and approaches 2000°C and even more at about 500 km depending upon solar activity.

5. Exosphere (बाह्यमंडल)

- "Exo" means outside.
- The exosphere is the outermost layer of the Earth's atmosphere.
- It starts at an altitude of about 500 km and goes out to about 10,000 km.
- The exosphere is largely composed of light gases such as hydrogen, carbon dioxide, atomic oxygen, and helium.





Scope and importance of environment

(पर्यावरण का क्षेत्र और महत्व)

1. Ecological Balance

- Maintenance of the equilibrium between living organisms and their environment.
- Supports biodiversity, ensures the survival of species, and maintains natural processes.

2. Human Health

- The impact of the environment on physical and mental well-being.
- Clean air, water, and soil are essential for good health. Pollution can lead to diseases.

3. Economic Importance

- The role of natural resources and ecosystem services in the economy.
- Provides raw materials, supports industries like agriculture, tourism, and fisheries, and offers ecosystem services like pollination and climate regulation.

4. Cultural and Aesthetic Value

- The significance of the environment in cultural heritage, recreation, and aesthetics.
- Natural landscapes inspire art, provide recreational opportunities, and hold cultural significance.

5. Sustainability

- The capacity to maintain ecological and human systems over the long term.
- Ensures resources are available for future generations, promotes conservation, and supports sustainable development.

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NEED FOR PUBLIC AWARENESS

(जन जागरूकता की आवश्यकता)

1 Environmental Education

- Integrating environmental topics in school curriculums.
- Community education programs.

2 Policy and Legislation

- Understanding and supporting environmental laws.
- Active participation in policy-making processes.

3 Sustainable Practices

- Encouraging recycling, waste reduction, and sustainable consumption.
- Promoting renewable energy sources.

4 Global Cooperation

- Importance of international agreements (e.g., Paris Agreement).
- Role of global organizations (e.g., UNEP).

5 Individual Actions

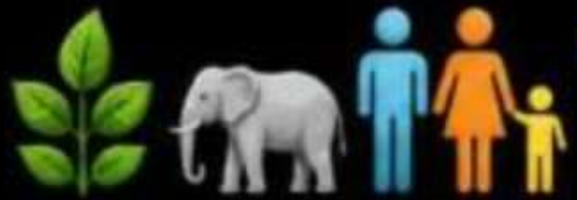
- Personal responsibility in reducing carbon footprints.
- Conservation efforts at the household level.

Ecosystem (पर्यावरण तंत्र)

"The system resulting from the integration of all living and non-living factors of the environment".

"A system formed by the interaction of all living organisms with each other and with physical and chemical factors of the environment in which they live, all linked by the transfer of energy and material.".

All plants, animals, and human beings



→ Depends on their immediate surroundings.

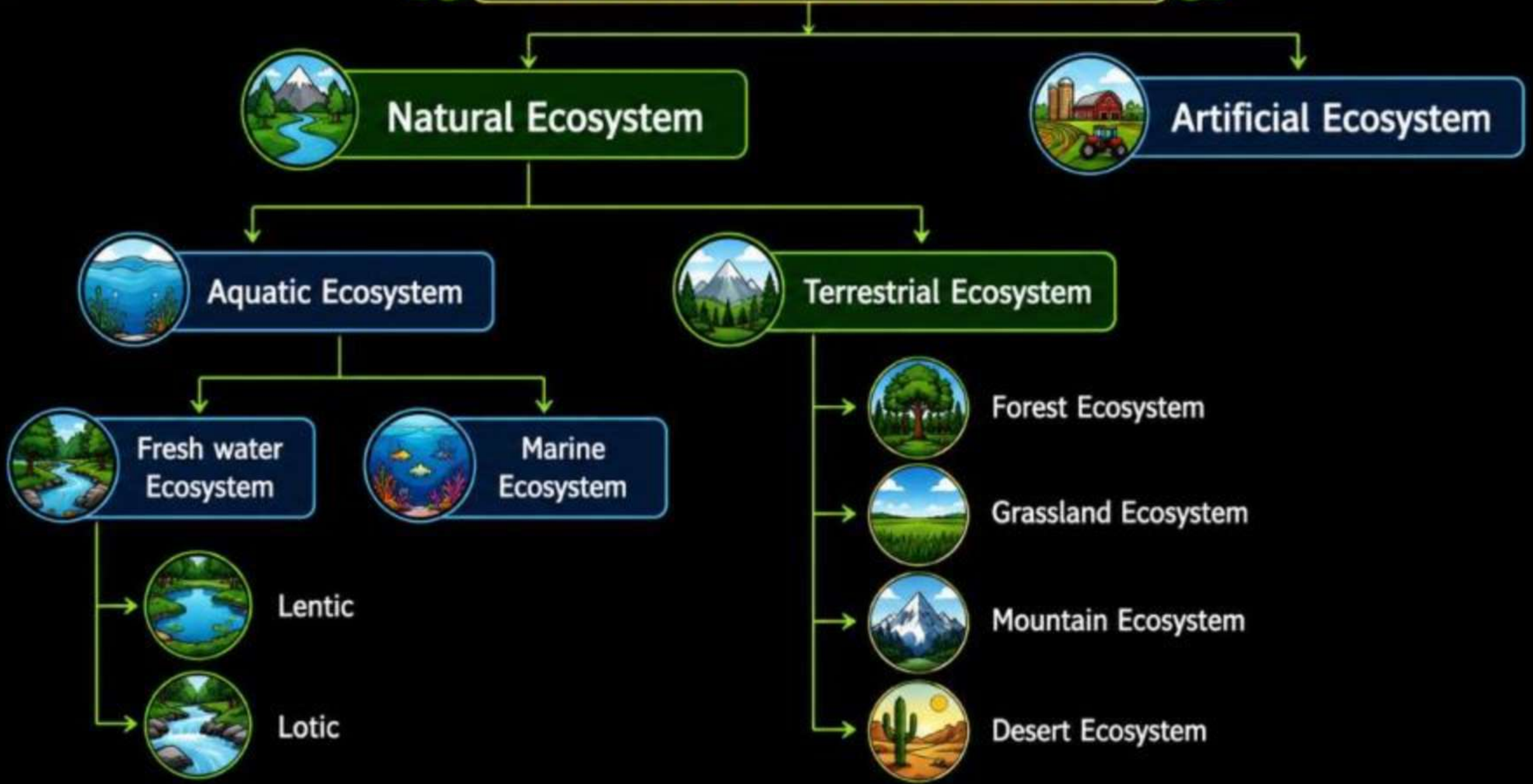
→ Relation between the living organisms and their surroundings forms an ecosystem.

→ Example

Ecosystem → Example → Rain, forest, Grassland, Dessert, Lake, Pond etc.

Ecology → Study of ecosystem → A branch of science which concerned with the inter-relationship of organism and their environment.

TYPES OF ECOSYSTEM



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TYPES OF ECOSYSTEM

Ecosystem are of two types -

(A) NATURAL ECOSYSTEMS

- These operate by themselves under natural conditions without any major interference by man. These are further divided as-

a) Terrestrial Ecosystem

- Found on Land : Forest, Grassland, Desert ecosystem etc.

Forest Grassland Desert Mountain

b) Aquatic Ecosystem

- It is divided into –

(i) Fresh Water - It may be Lotic (Running water as spring, stream or rivers) or Lentic (Standing water as lake, ponds, pools etc).

Lotic (Running water)

Lentic (Standing water)

(ii) Marine Water - Like Ocean and seas.

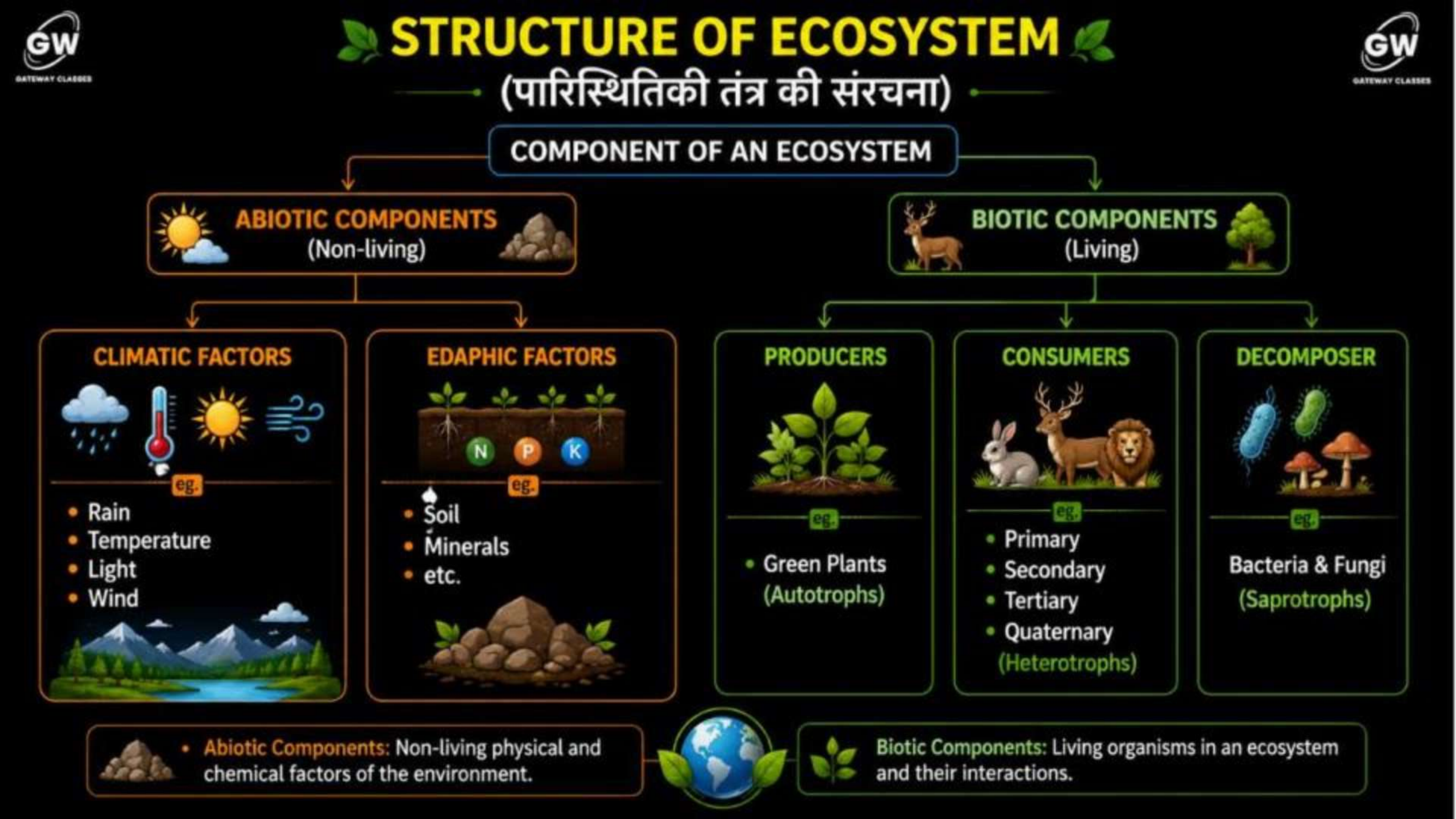
Marine Water (Ocean & Seas)

(B) ARTIFICIAL ECOSYSTEM

- It is the ecosystem in which conditions are modified by man.

Examples- Aquarium, Zoo, Cultivated land, Fish farms, Canals, Gardens and Industrial & Urban area.

Aquarium Zoo Cultivated land Fish farms Canals Industrial & Urban area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GATEWAY CLASSES
STRUCTURE OF ECOSYSTEM
(पारिस्थितिकी तंत्र की संरचना)
COMPONENT OF AN ECOSYSTEM
ABIOTIC COMPONENTS
(Non-living)
BIOTIC COMPONENTS
(Living)
CLIMATIC FACTORS
eg.
• Rain
• Temperature
• Light
• Wind
EDAPHIC FACTORS
N
P
K
eg.
• Soil
• Minerals
• etc.
PRODUCERS
eg.
• Green Plants
(Autotrophs)
CONSUMERS
eg.
• Primary
• Secondary
• Tertiary
• Quaternary
(Heterotrophs)
DECOMPOSER
eg.
Bacteria & Fungi
(Saprotrophs)
• Abiotic Components: Non-living physical and chemical factors of the environment.
Biotic Components: Living organisms in an ecosystem and their interactions.
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STRUCTURE OF ECOSYSTEM

An ecosystem may be divided into two components.

1. ABIOTIC COMPONENTS (NON LIVING)

These are the non living components of an ecosystem includes both physical and chemical features.

PHYSICAL FEATURES → Wind, Soil, Moisture, Temperature, Light

Physical features includes wind, soil, moisture, temperature, light etc.

CHEMICAL FEATURES → Water, Oxygen (O_2), Minerals, Iron (Fe), Sulfur (S), Carbon (C), Nitrogen (N)

Chemical features includes water, gases as oxygen, minerals as iron, sulfur, carbon, nitrogen etc.

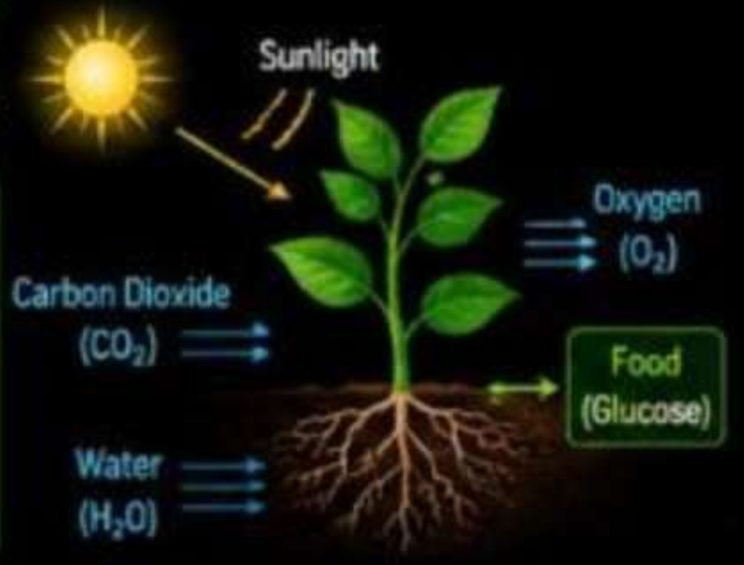
The kind of chemicals which are present in any ecosystem may regulate the activities of autotrophs and heterotrophs.

2. BIOTIC COMPONENTS (LIVING)

The biotic components of an ecosystem may be divided as

(A) AUTOTROPHS OR PRODUCERS
(स्वपोषी या उत्पादक)

- These are the living components of an ecosystem which can prepare their own food.
- These are mainly green plants which prepare their food in presence of sunlight by making use of carbon dioxide, water and chlorophyll through the process of photosynthesis.



STRUCTURE OF ECOSYSTEM

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(B) HETEROTROPHS CONSUMERS (उपभोक्ता)

These are organisms which feed upon green plants (autotrophs). They are divided as herbivores, carnivores and omnivores.

(i) HERBIVORES (शाकाहारी)

- Those organisms which feed on green plants only are called herbivores.
- They are also called as primary consumers.

Examples

Cow Buffalo Goat Deer Camel

(ii) CARNIVORES (मांसाहारी)

- Those organisms which feed on meat are called carnivores.
- They are also called as secondary consumers.

Examples

Lion Tiger Wolf

(iii) OMNIVORES (सर्वाहारी)

- Those organisms which feed on green plants as well as meat are called omnivores.
- They are also called as tertiary consumers.

Examples

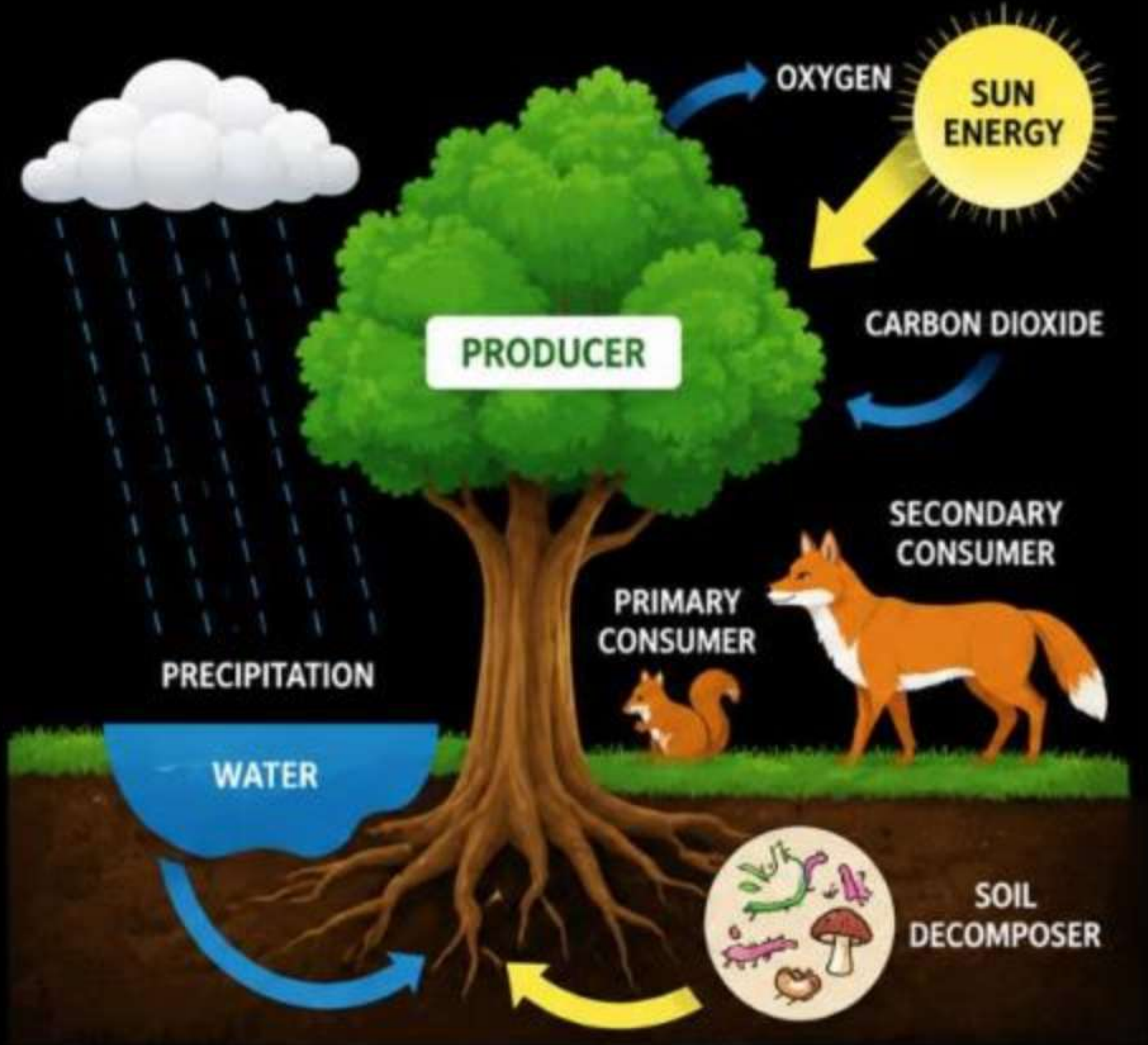
Human beings Dog

Herbivores are primary consumers, carnivores are secondary consumers and omnivores are tertiary consumers in an ecosystem.

(C) Decomposers (अपघटक)

→ Those organisms which feeds on dead bodies of consumers and decompose the complex organic compounds into simple organic compounds and make them available for autotrophs (producers).

→ They includes mainly bacteria and fungi.

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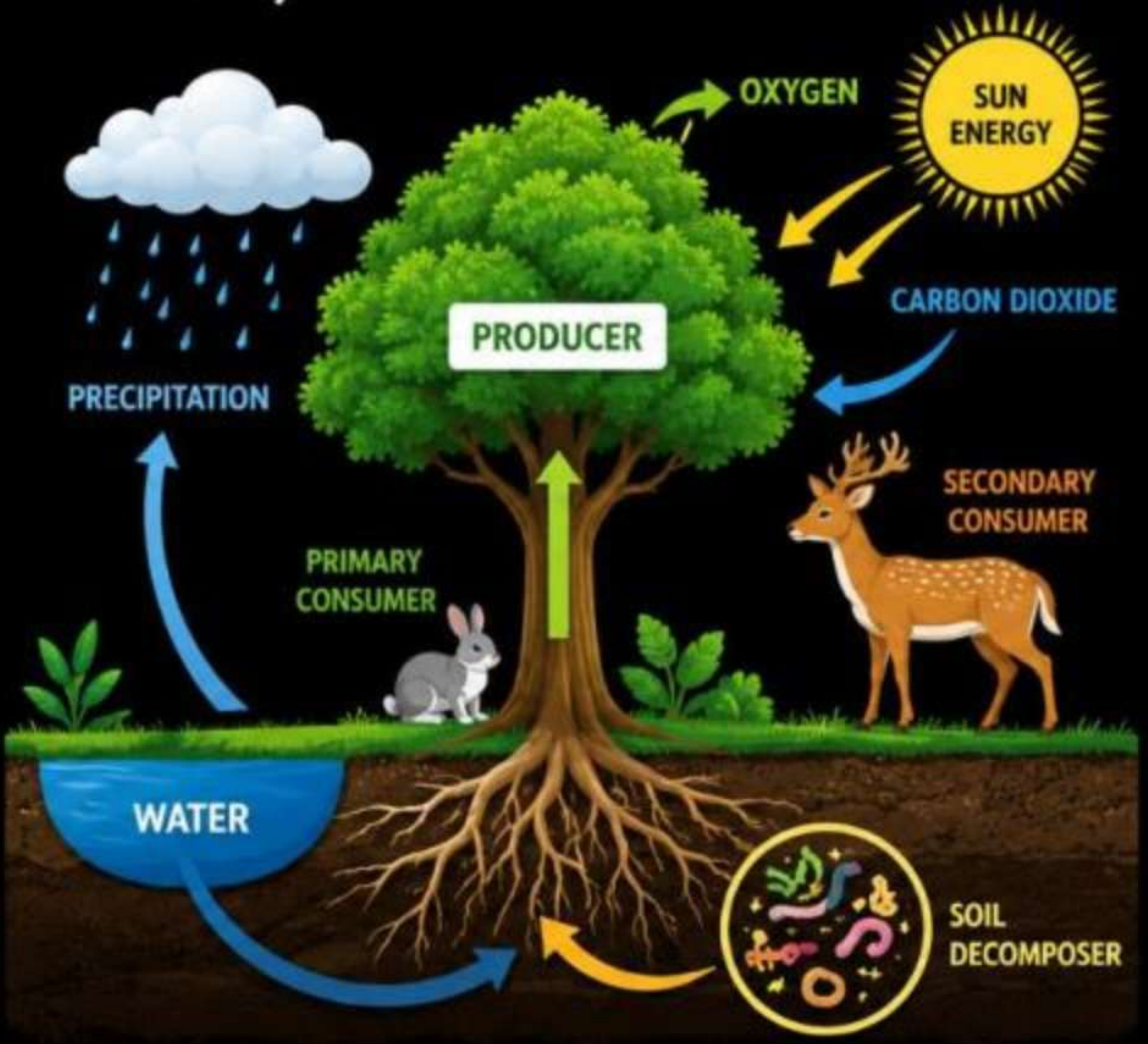
FUNCTIONS OF ECOSYSTEM
(पारिस्थितिकी तंत्र के कार्य)

→ The functions of an ecosystem are essential for maintaining the balance of the environment and supporting life.

These are the following functions of ecosystem-

- Energy Flow
- Nutrient Cycling
- Regulation of Ecological Processes (climate, water purification, and disease control)
- Habitat Provision
- Soil Formation and Maintenance



... ENERGY FLOW IN AN ECOSYSTEM

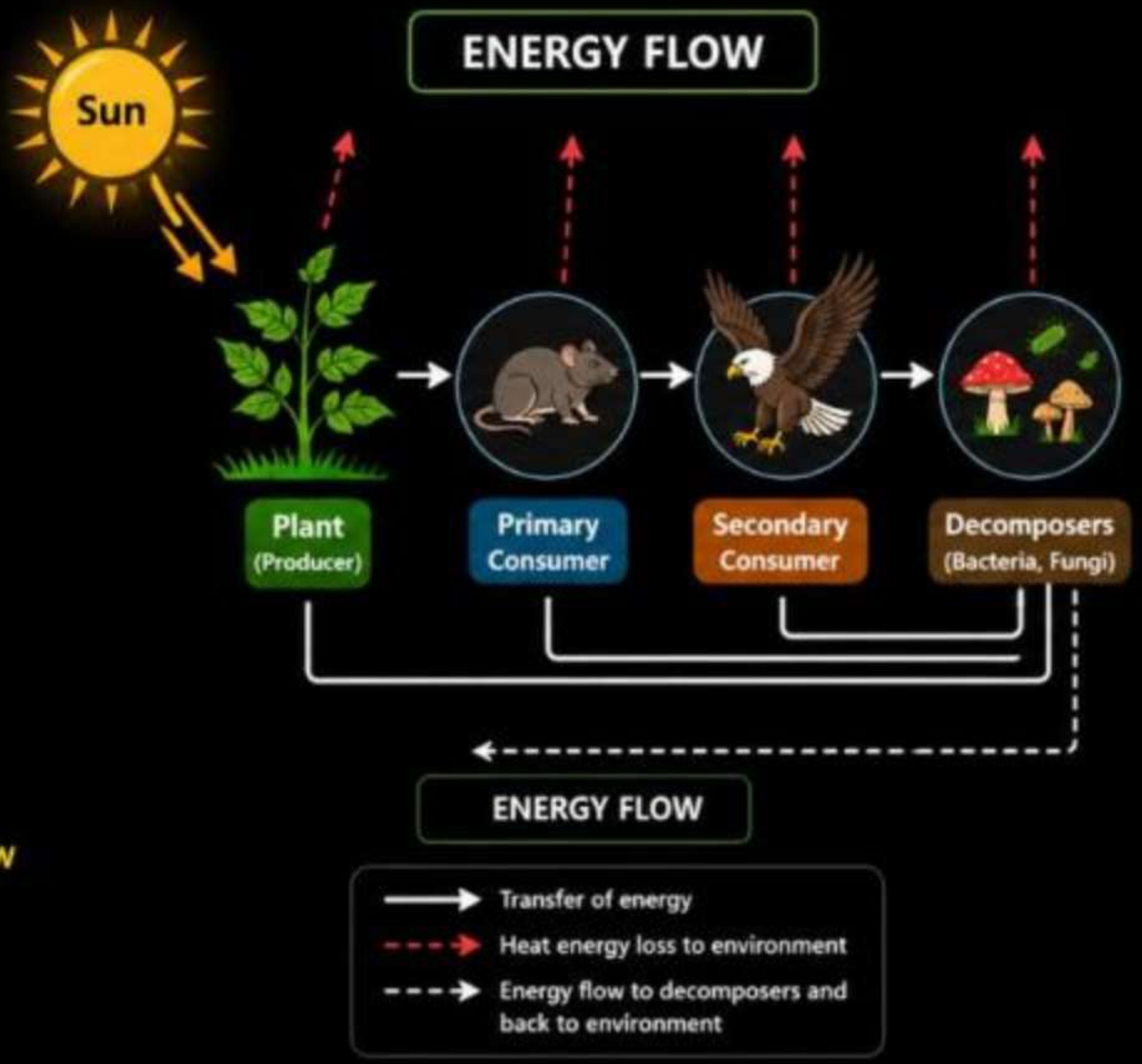
→ Energy flow is the movement of energy through an ecosystem through a series of organisms and back to the external environment.

→ According to first law of thermodynamics, energy can neither be created nor destroyed; energy can simply change its form.

→ In an ecosystem, autotrophs i.e. green plants fix the radiant energy of the sun and converted it into chemical energy.

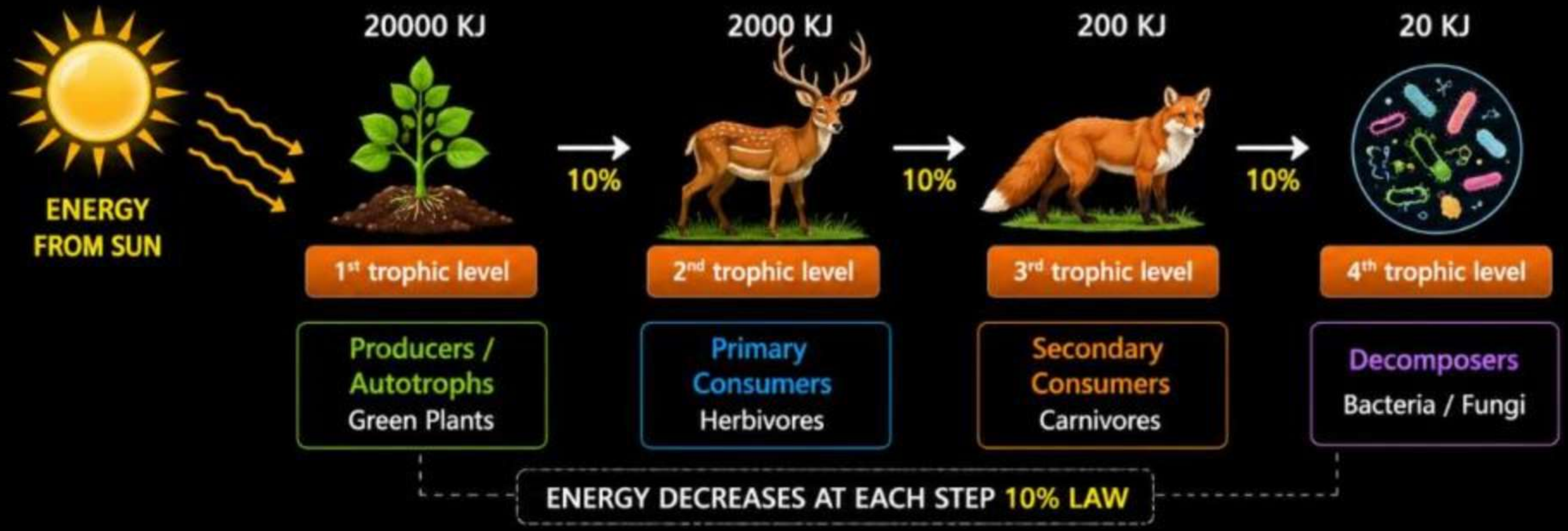
→ This chemical energy is transferred to heterotrophs i.e. consumers when they feed upon autotrophs and finally when heterotrophs die this energy is transferred to decomposers.

→ Hence in this process of energy flow, we can say that the first law of thermodynamics is applicable in living organisms.



ENERGY FLOW IN AN ECOSYSTEM

- According to the second law of thermodynamics, during energy transformation, large part of energy is lost in the form of **heat** into the atmosphere.
- As shown in figure, there is a **successive decrease** in energy at higher trophic levels.
- The Ten percent law of energy flow states that when the energy is passed on, from one trophic level to another, only **10% of the energy** is passed on.



FOOD CHAIN (खाद श्रृंखला)

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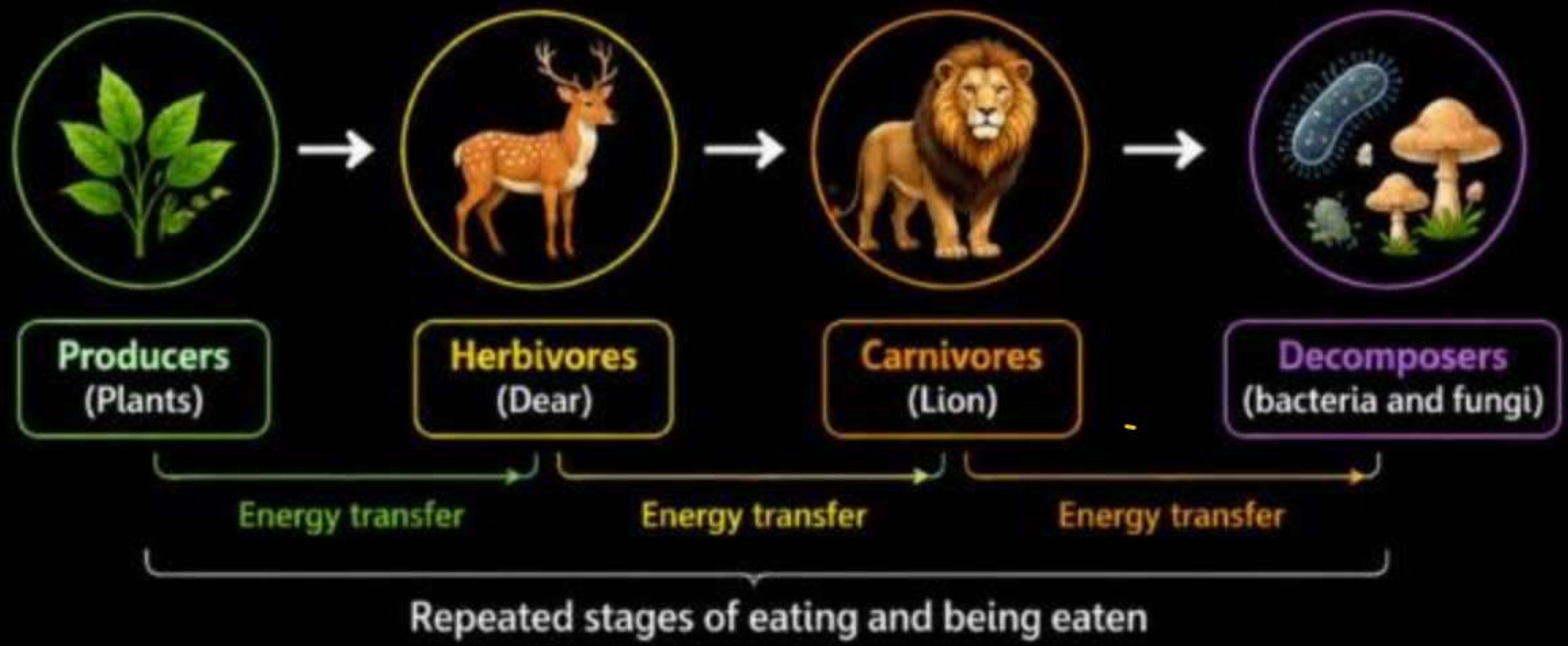
- The transfer of food energy from the source in plants (producers) through a series of organisms (herbivores → carnivores → decomposers) with repeated stages of eating and knowg eaten is known as the food chain.
- In any food chain, energy (in the form of food) flow from producers to primary consumers (herbivores), from primary consumers to secondary consumers (carnivores), from secondary consumers to tertiary consumers (carnivores/omnivores) and so on.
- **The transfer of energy from one trophic level to the next trophic level is called food chain.**

There are three types of food chain:

1. Grazing Food Chain
2. Detritus Food Chain
3. Parasitic Food Chain



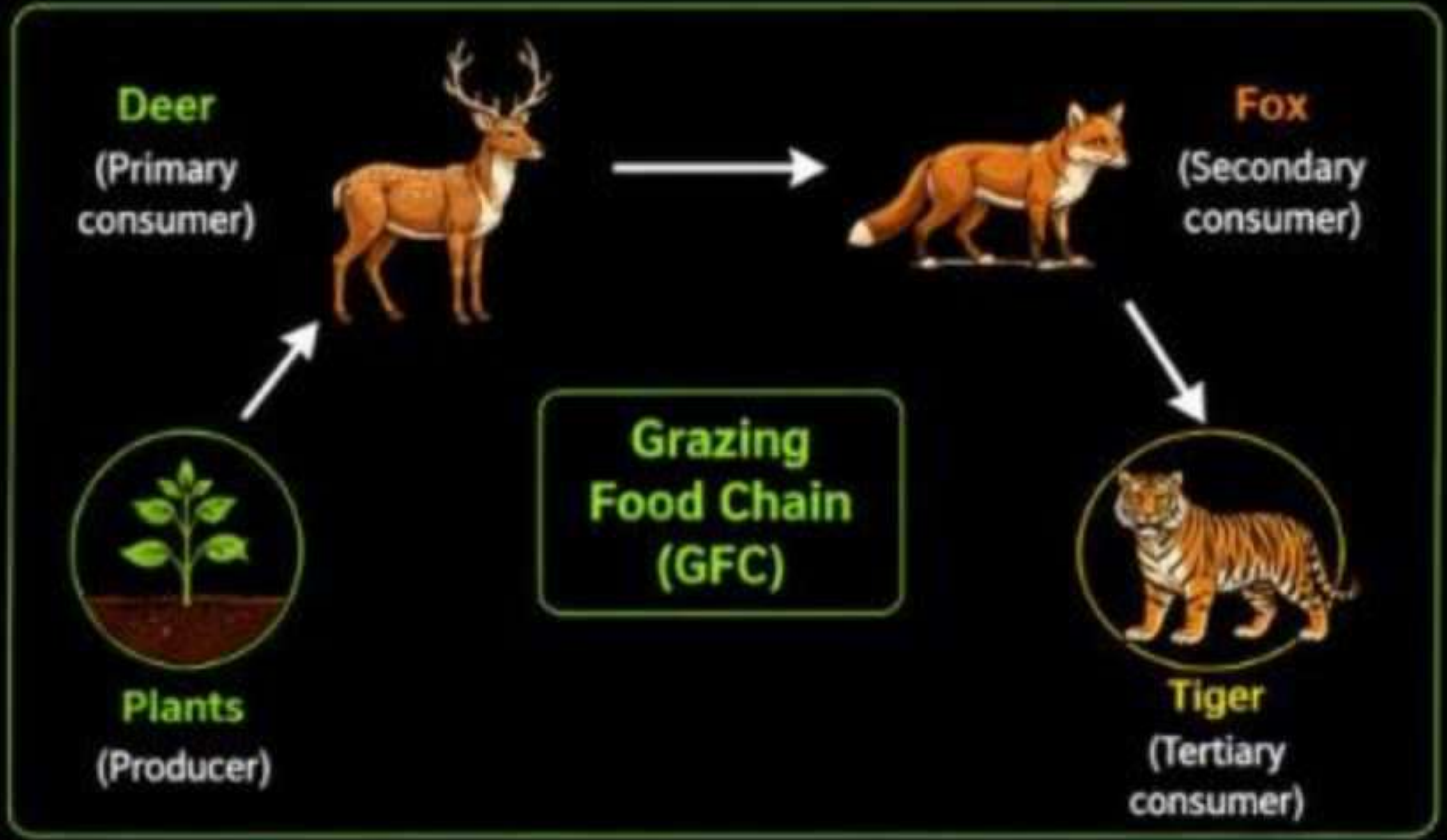
1. GRAZING FOOD CHAIN

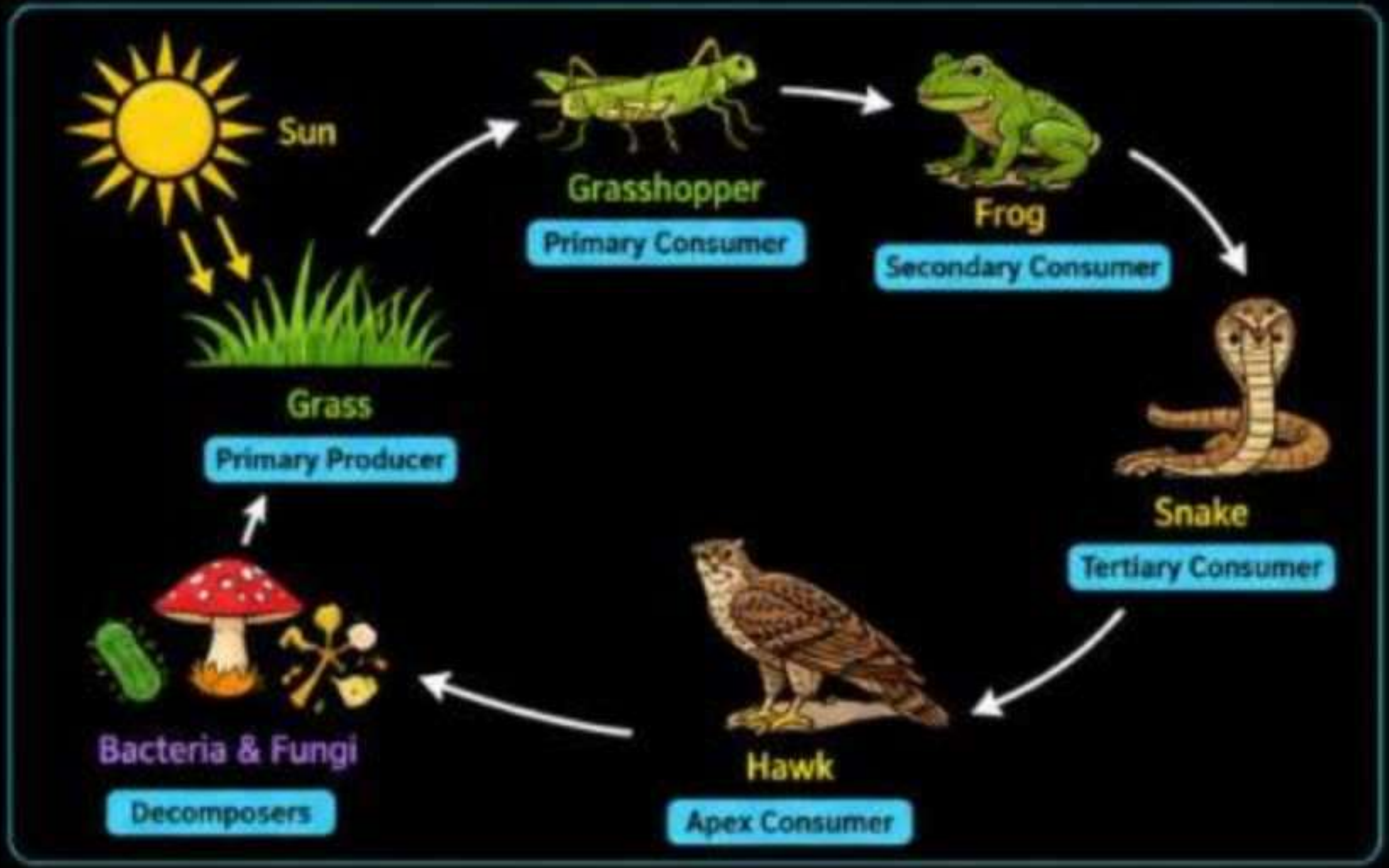
- This type of food chain starts from the living green plants goes to grazing herbivores, and on to carnivores.
- Ecosystems with such type of food chain are directly dependent on an influx of solar radiation.

EXAMPLES ARE

1. Green plants → Deer → Tiger
2. Grasses → Grasshopper → Frog → Snake → Hawk

Decomposers





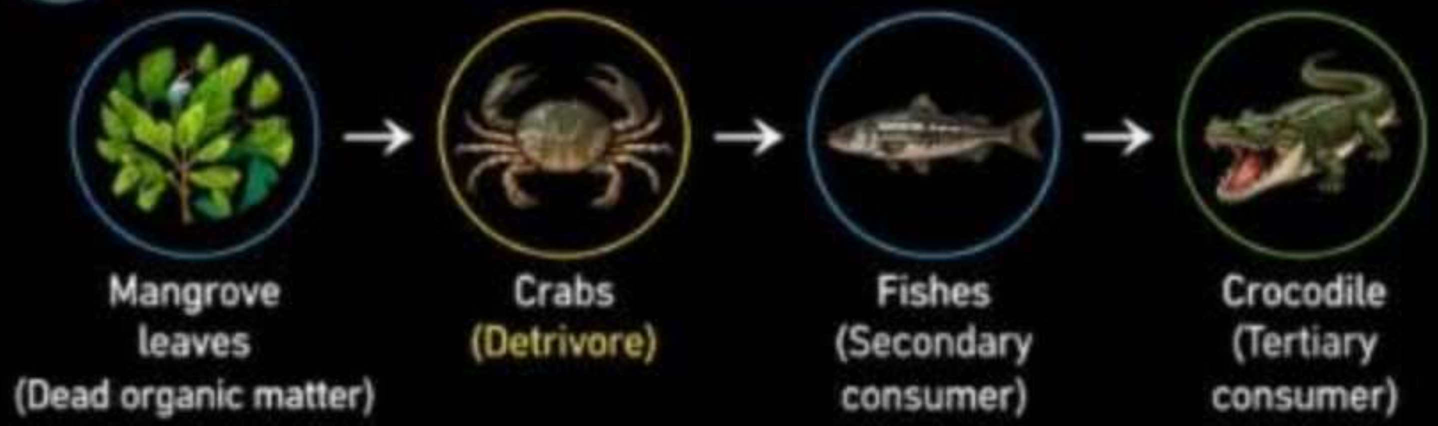
2. DETRITUS FOOD CHAIN

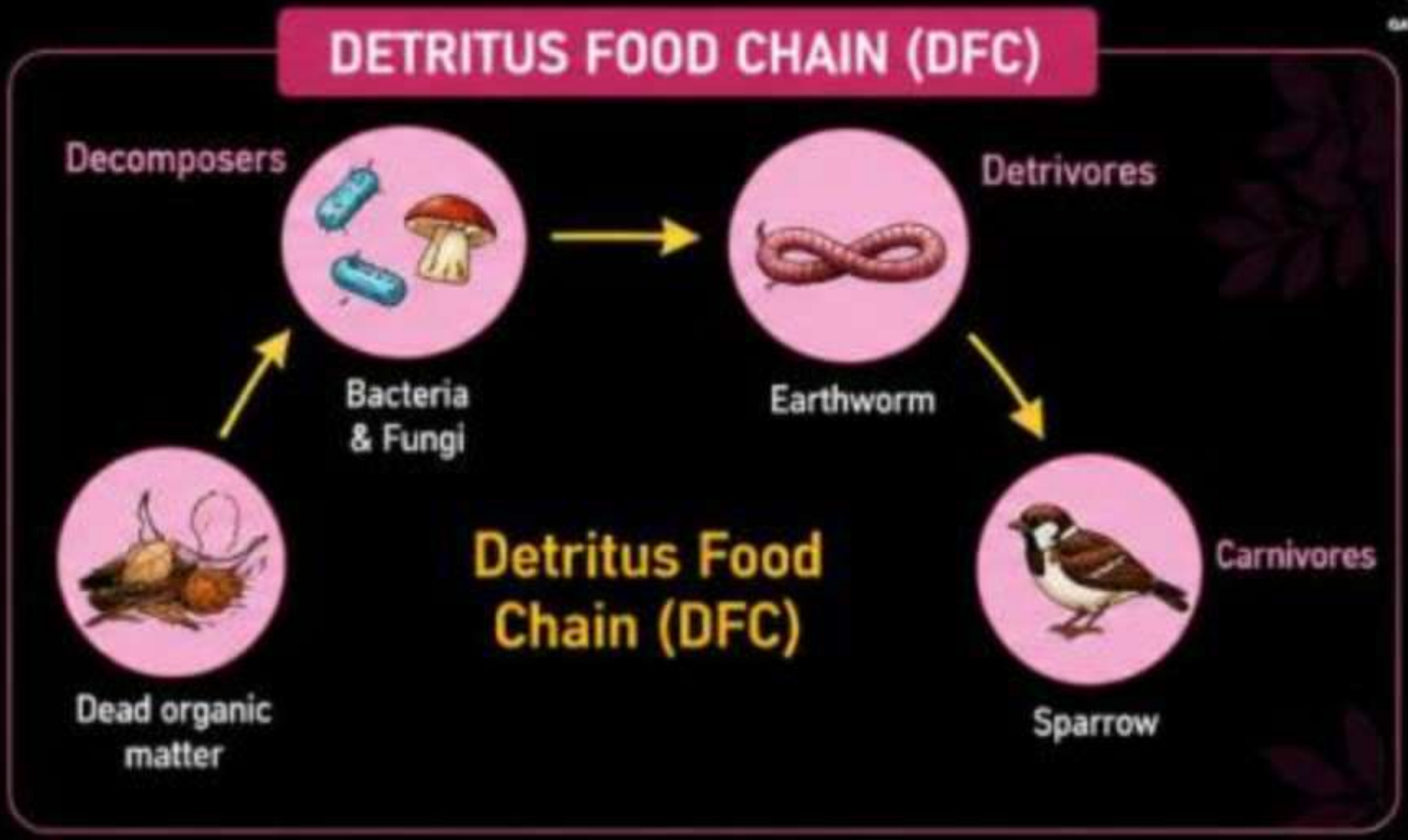
- It starts from dead organic matter and goes to detritus feeding organisms (**detritivores**) and on their predators.
- **Detrivores** are consumers of dead organic matter.

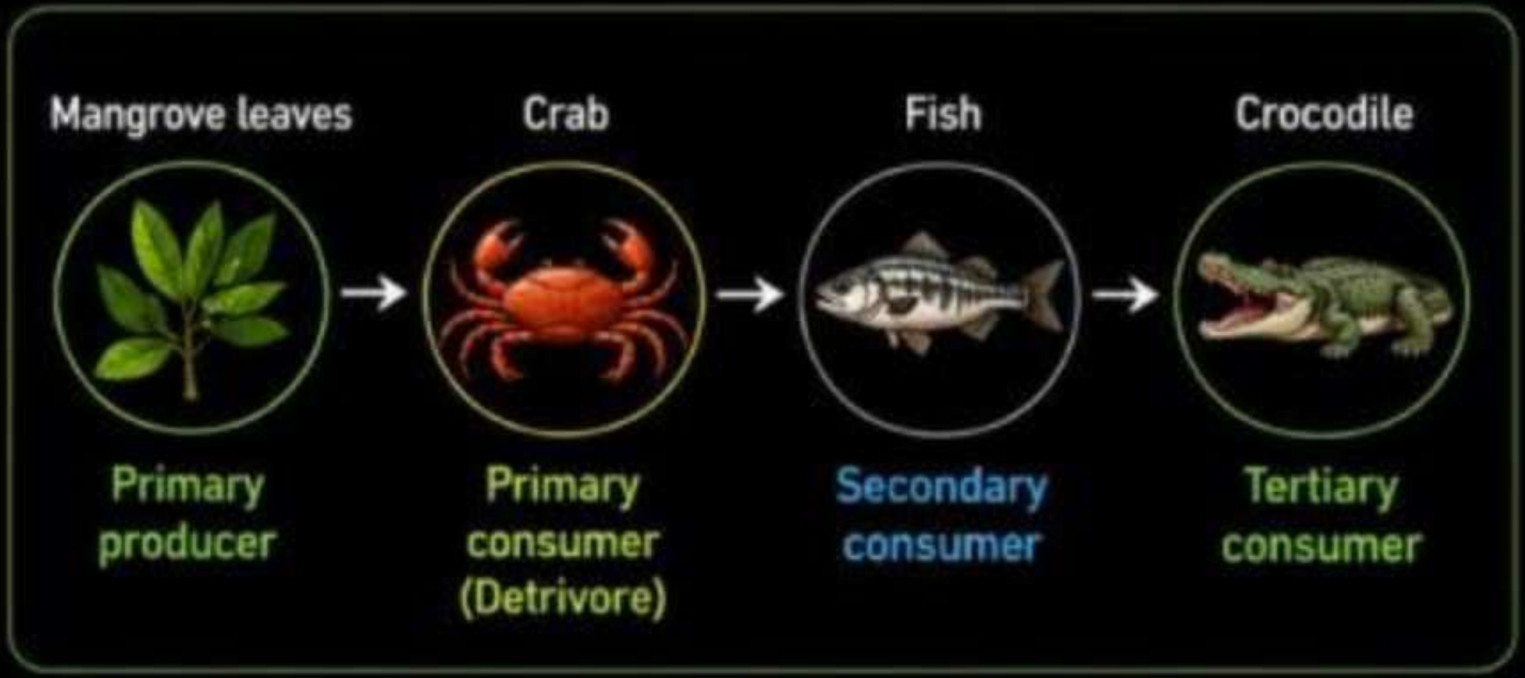
EXAMPLES ARE

1. Dead organic matter → Detrivores → Predators

2. Mangrove leaves → Crabs → Fishes → Crocodile

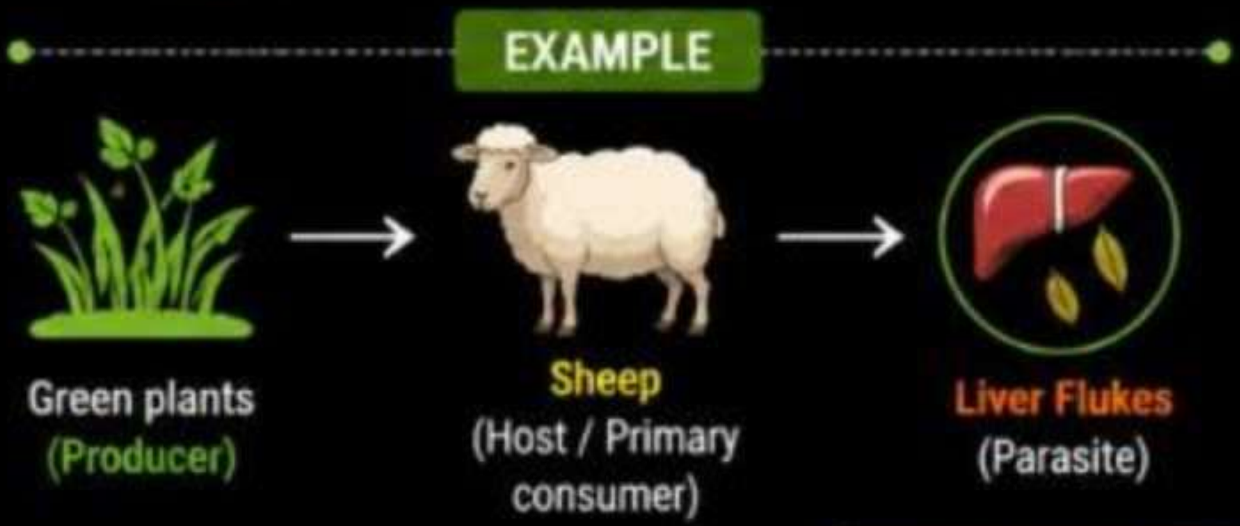


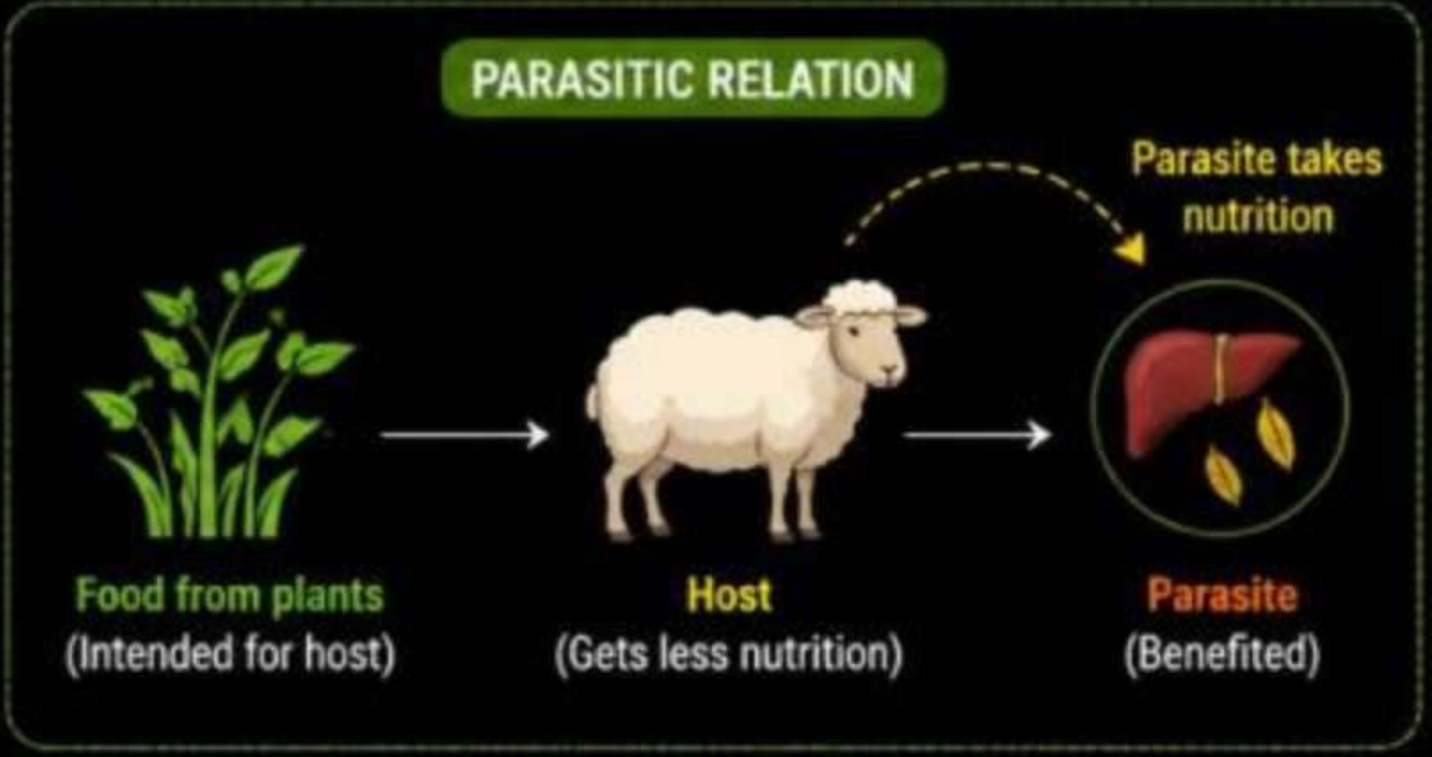




3. PARASITIC FOOD CHAIN

- In such type of food chain, **parasites** may consume a portion of food available to the host.
- In the whole process the **parasite** gets nutritionally benefited and **host** is harmed.





SIGNIFICANCE OF FOOD CHAIN

- **Energy Transfer:** This shows how energy moves through different trophic levels.
- **Nutrient Cycling:** Illustrates the flow of nutrients in ecosystems.
- **Ecological Balance:** Helps maintain stability and balance in ecosystems.
- **Understanding Ecosystems:** Provides insights into species interactions and dependencies.
- **Conservation Efforts:** Aids in identifying critical species and managing ecosystems.
- **Environmental Health Indicator:** Reflects changes and impacts on ecosystems.

FOOD WEB (खाद्य जाल)

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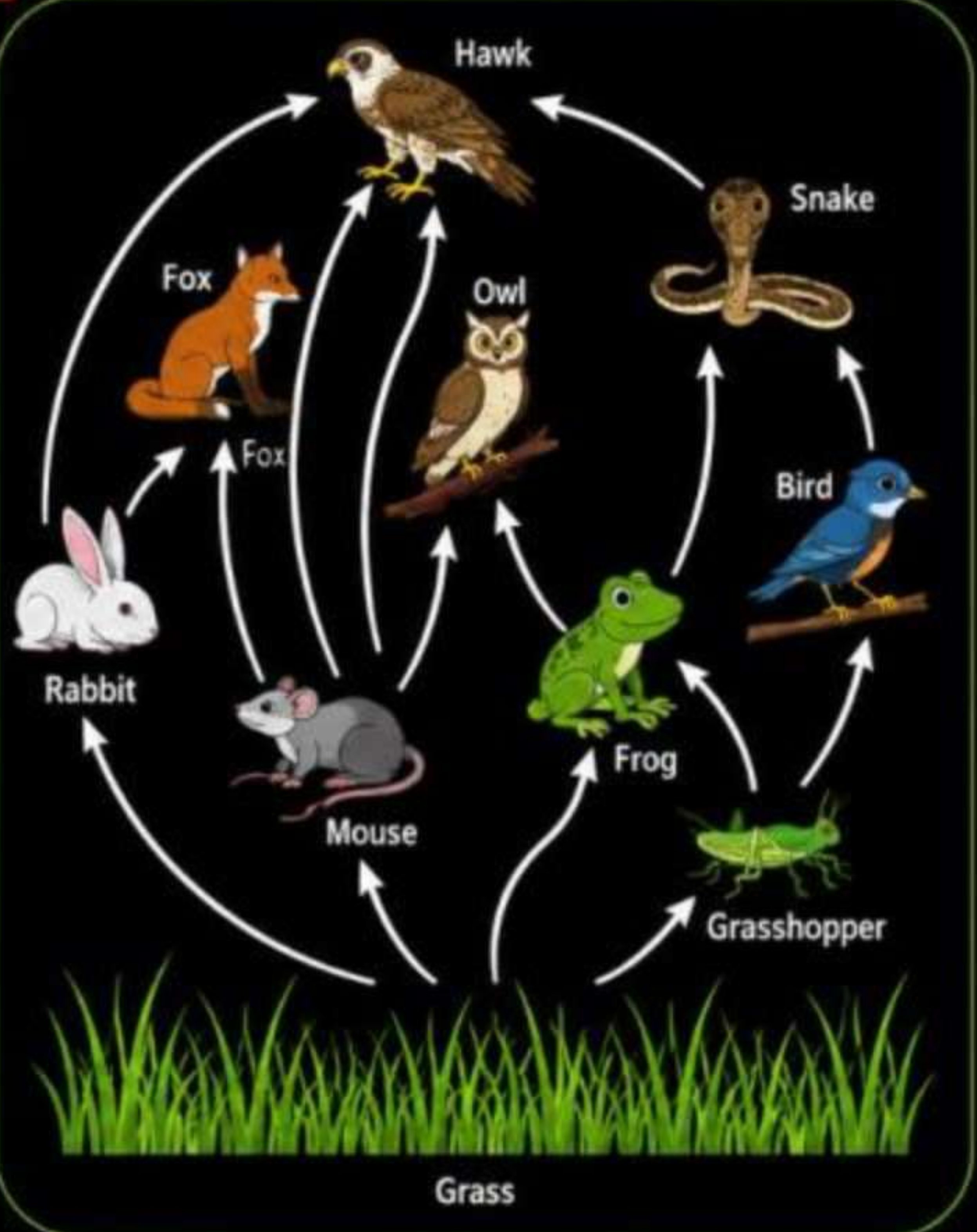
- A food web is a tool that illustrates the feeding relationships among species within a specific habitat.
- Species may occupy different trophic levels in an ecosystem.
- For example, a single plant may be eaten by more than one species for their nourishment, herbivores feed on different kinds of plants. Similarly, carnivores may feed on different kinds of herbivores, and so on.
- Therefore, we can say that food chains are interconnected in various ways.
- **Such types of interconnected food chains form a web called a food web."**

- A network of food chains which are interconnected at various trophic levels, so as to form a number of feeding connections amongst different organisms of a biotic community is called food web".



1. Pyramid of Numbers

→ There is a gradual decrease in number of individuals from lower trophic level to higher trophic levels.

→ In a **forest** or **grassland** ecosystem autotrophs support a lesser number of herbivores which support fewer carnivores.

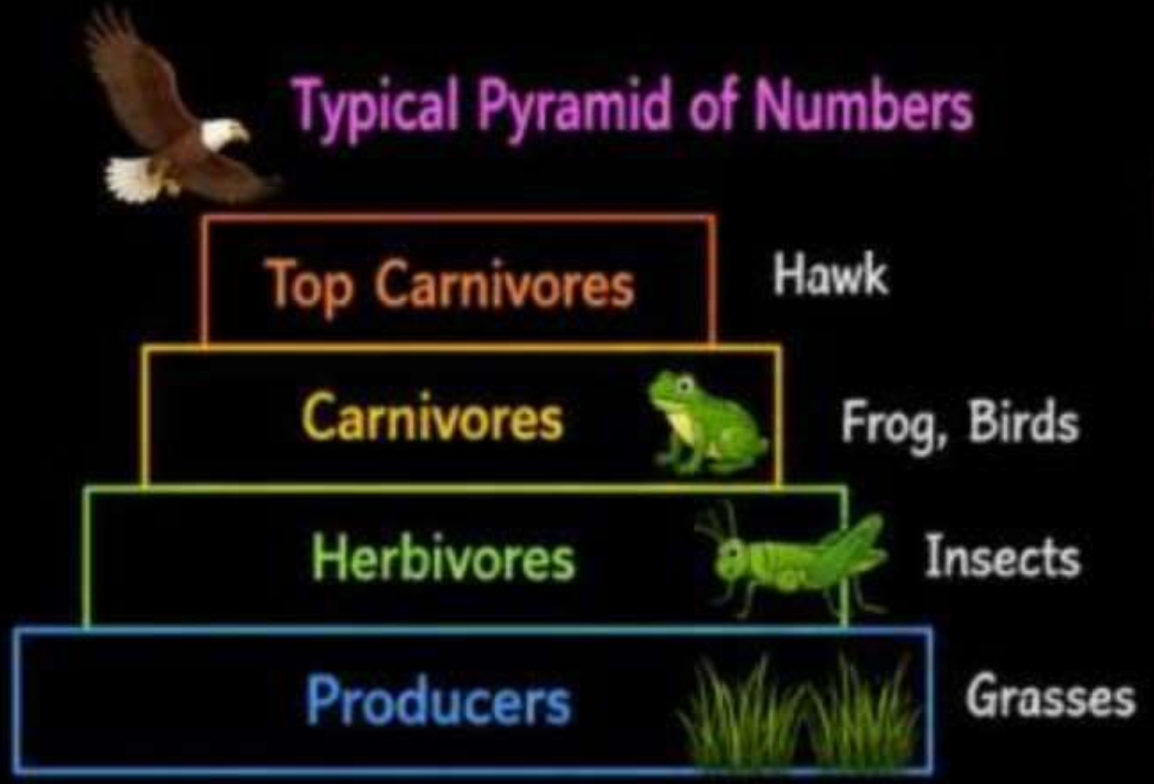


Fig. Grassland

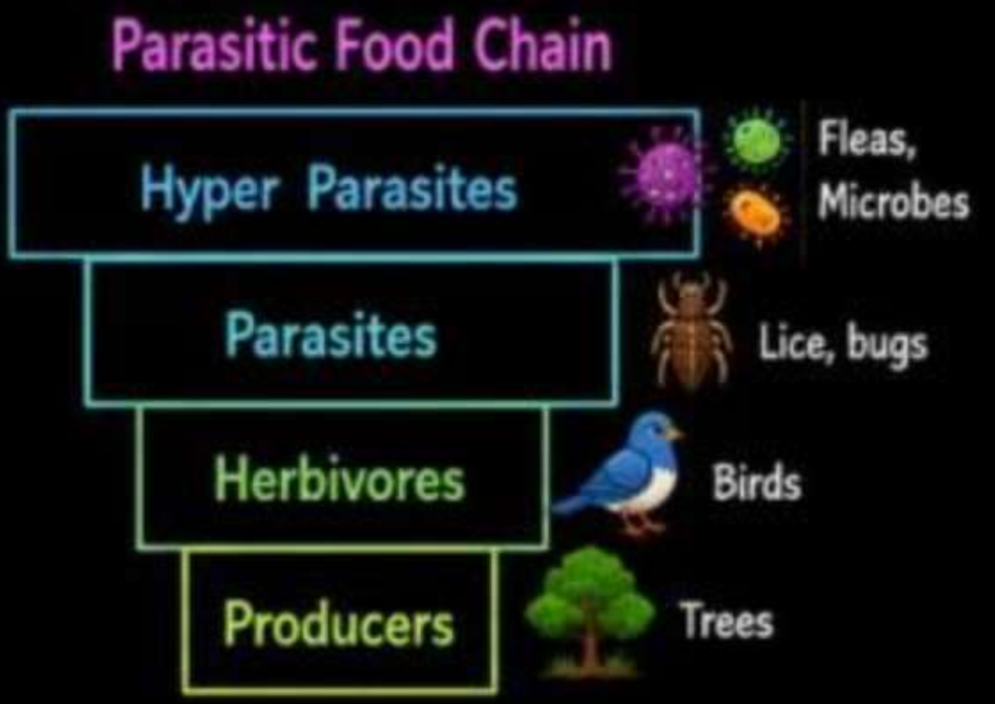


Fig. Parasitic food chain

→ In case of parasitic food chain, single tree may support variety of herbivores and these herbivores support various parasites.

→ These parasites may support varieties of hyper parasites like bacteria and fungi.

→ Therefore in **parasitic or inverted pyramid** there is a gradual increase in the number as we go up from lower to higher trophic level.

2. Pyramid of Biomass

→ In the pyramid of biomass, there is a gradual **decrease** in biomass from lower to higher trophic level i.e. a decrease in biomass from producers to primary consumers and so on.

→ In other words the maximum biomass may be formed in producers and the minimum in tertiary as top consumers.

→ In an inverted pyramid of biomass, there is a gradual **increase** in the biomass as we go up from lower to higher trophic levels.

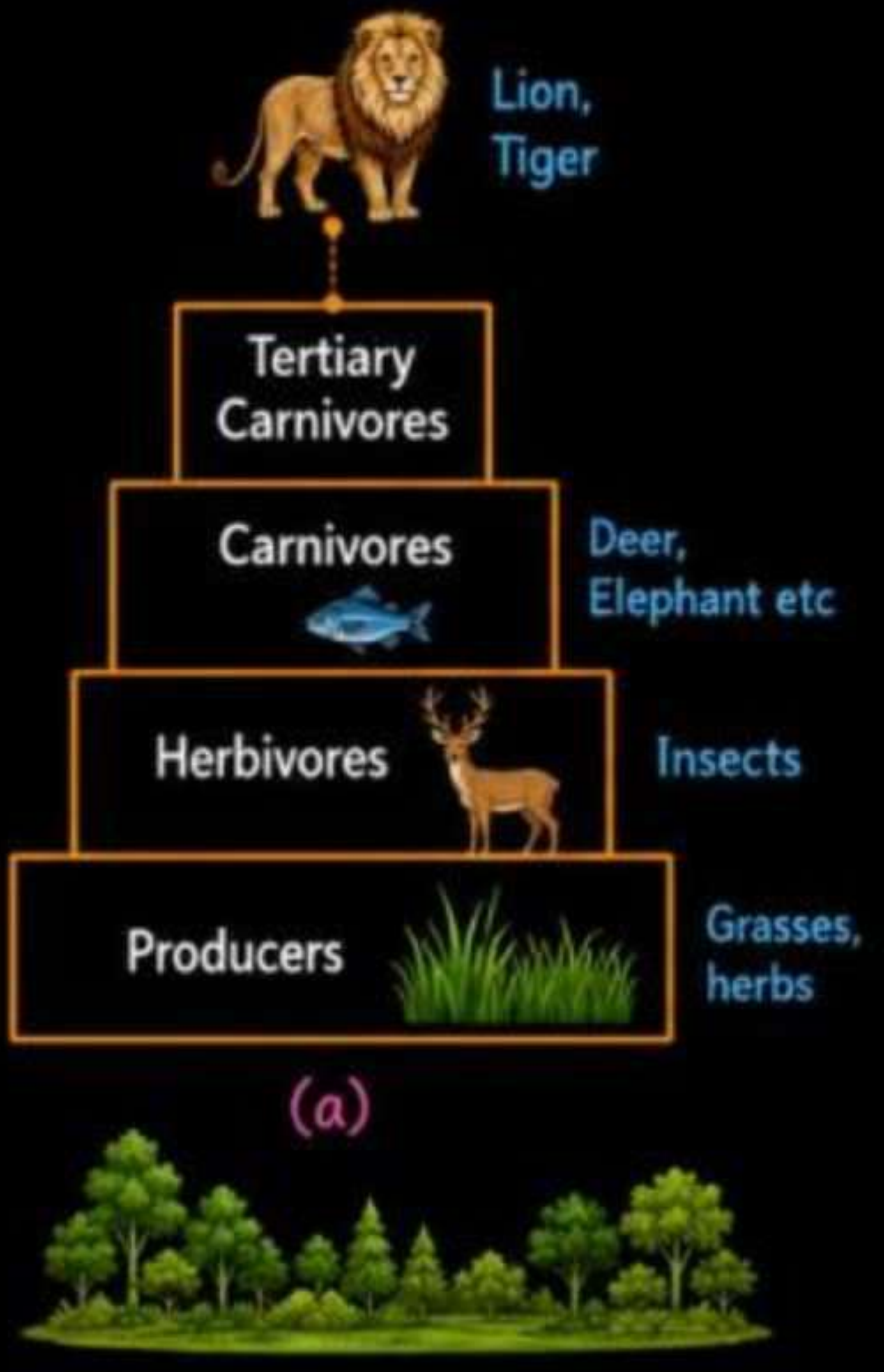


Fig. Pyramid of biomass
(a) Forest

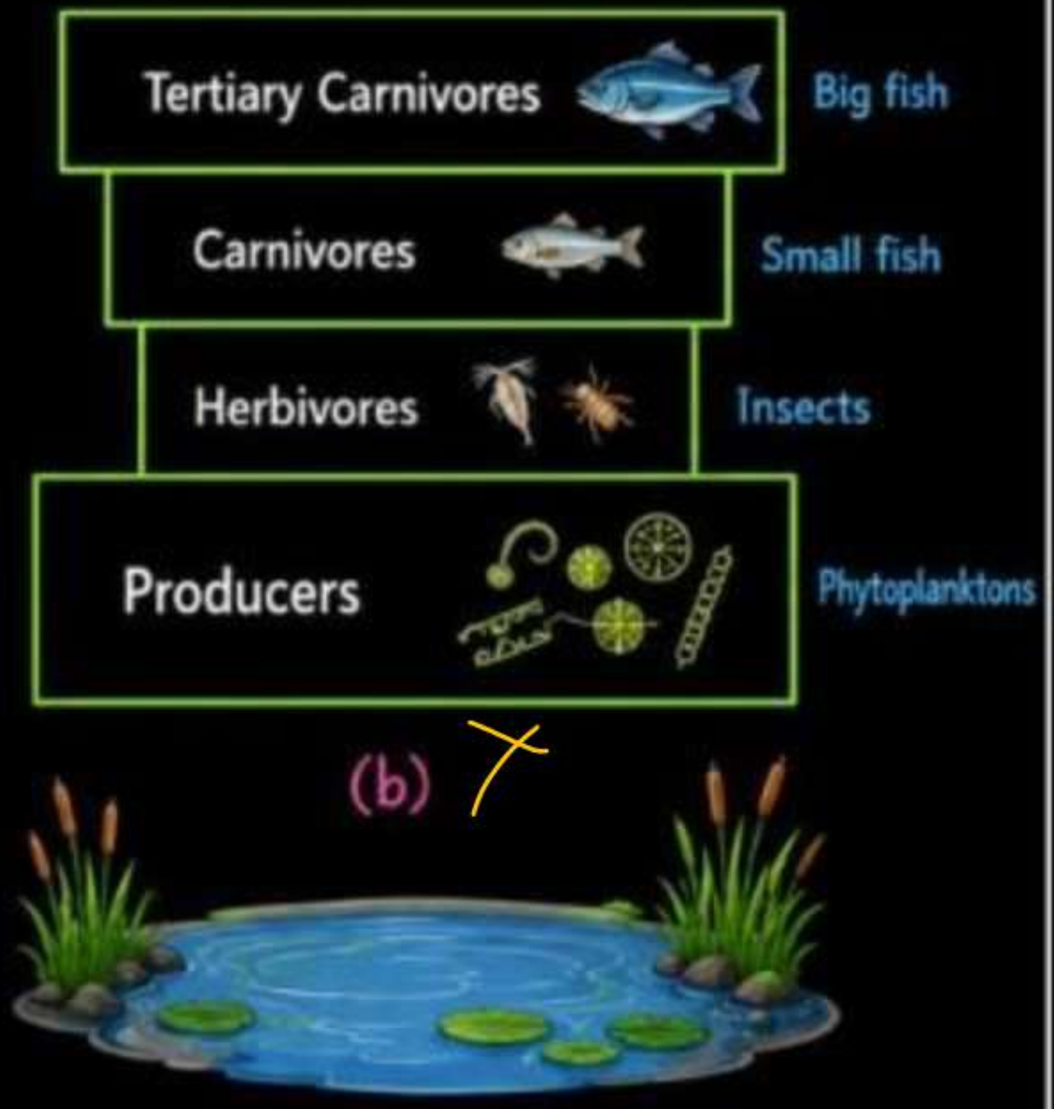


Fig. Pyramid of biomass
(b) Pond

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3. Pyramid of Energy

→ In pyramid of energy, there is a gradual decrease in energy as we go up from lower trophic level to higher tropic level.

→ According to second law of thermodynamics there is considerable loss of energy in the form of heat as we go up higher in trophic level.

→ More energy is available in autotrophs in comparison to primary consumers.

→ Therefore we can easily conclude that shorter the food chain higher will be the amount of energy at top.

→ The pyramid of energy is always upright due to unidirectional flow of energy in food chain.

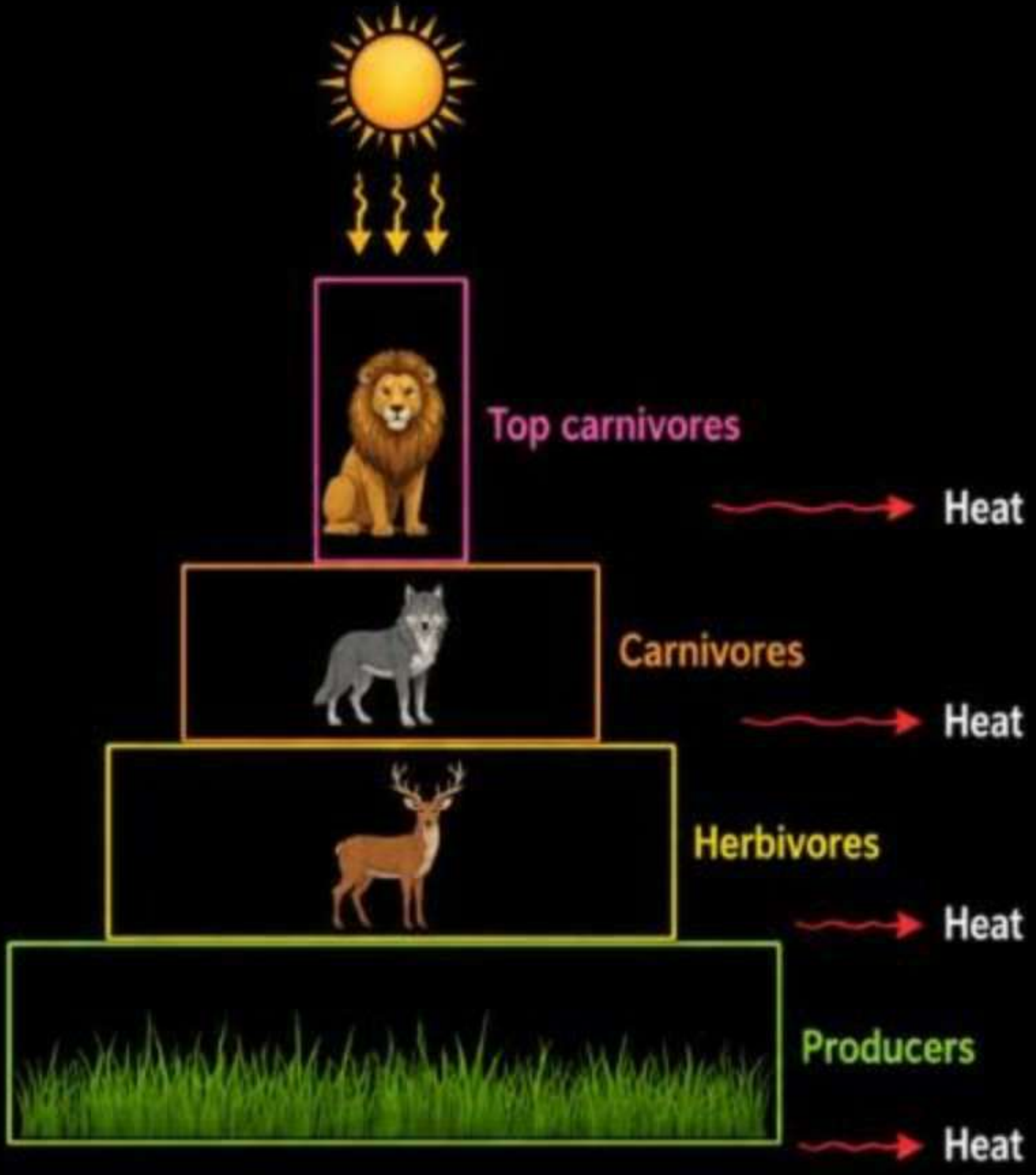
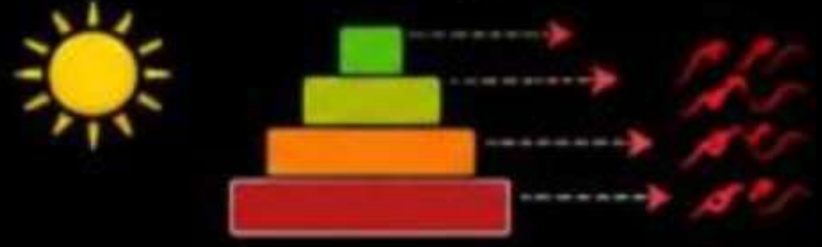


Fig: Pyramid of energy

Why are there fewer Organisms at the Top of the Ecological Pyramid?

1. Energy Loss at Each Trophic Level



2. Smaller Energy Base

3. Greater Energy Needs of Higher Trophic Levels

Importance of Ecological Pyramid

The importance of the ecological pyramid can be explained in the following points:

Energy Flow: Illustrates the decrease in energy as it moves up trophic levels.

Ecosystem Structure: Visual representation of trophic levels and species interactions.

Biomass Distribution: Shows the amount of biomass at each trophic level.

Population Dynamics: Indicates population sizes and trends of different species.

Conservation and Management: Helps identify at-risk species and manage ecosystems sustainably.

Nutrient Cycling: Depicts how nutrients move through an ecosystem.

Ecological Efficiency: Demonstrates the efficiency of energy transfer in ecosystems.

SUSTAINABLE DEVELOPMENT (सतत विकास) (AKTU-2023-24,2017-18)

→ By 2050, our planet will need to support some ten billion people compared to about eight billion today.

→ This raises huge challenges in food, clothing, and shelter for this rapidly growing population.

→ Development has to be visualized in a holistic manner, where it brings benefits to all, not only for the present generation but also for future generations.

SUSTAINABLE DEVELOPMENT

> "The development that meets the needs of the present without compromising the ability of future generations to meet their own needs."

→ Sustainable development emphasizes on a balance between economic growth, environmental protection, and social equity.

Elements of Sustainable Development

1. Economic Sustainability:

→ Ensuring economic growth and development that creates wealth and employment, without depleting natural resources.

→ It focuses on improving the quality of life and reducing poverty, while maintaining economic activities over time.

2. Environmental Sustainability:

→ Protecting and managing natural resources and ecosystems to ensure that they can sustain future generations.

→ It involves reducing pollution, conserving biodiversity, and using resources efficiently to prevent degradation of the environment.

3 Social Sustainability:

- Promoting social equity, justice, and access to basic human rights and services.
- It ensures that development benefits all segments of society, including marginalized and disadvantaged groups, fostering inclusive communities.

4 Cultural Sustainability:

- Respecting and preserving cultural diversity and heritage.
- This ensures that development respects local customs, traditions, and values, while also promoting cultural exchange and mutual understanding.

SUSTAINABLE DEVELOPMENT GOALS (SDGs)

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→ The Sustainable Development Goals (SDGs) are a set of 17 global goals established by the United Nations in 2015, designed to be a blueprint for achieving a better and more sustainable future for all.

→ They address global challenges, including those related to poverty, inequality, climate change, environmental degradation, peace, and justice.

SUSTAINABLE DEVELOPMENT GOALS

THESE ARE THE FOLLOWING 17 SDGs:

1. No Poverty
2. Zero Hunger
3. Good Health and Well-Being
4. Quality Education
5. Gender Equality
6. Clean Water and Sanitation
7. Affordable and Clean Energy
8. Decent Work and Economic Growth
9. Industry, Innovation, and Infrastructure
10. Reduced Inequality
11. Sustainable Cities and Communities
12. Responsible Consumption and Production
13. Climate Action
14. Life Below Water
15. Life on Land
16. Peace, Justice, and Strong Institutions
17. Partnerships for the Goals

17 GOALS FOR A BETTER TOMORROW

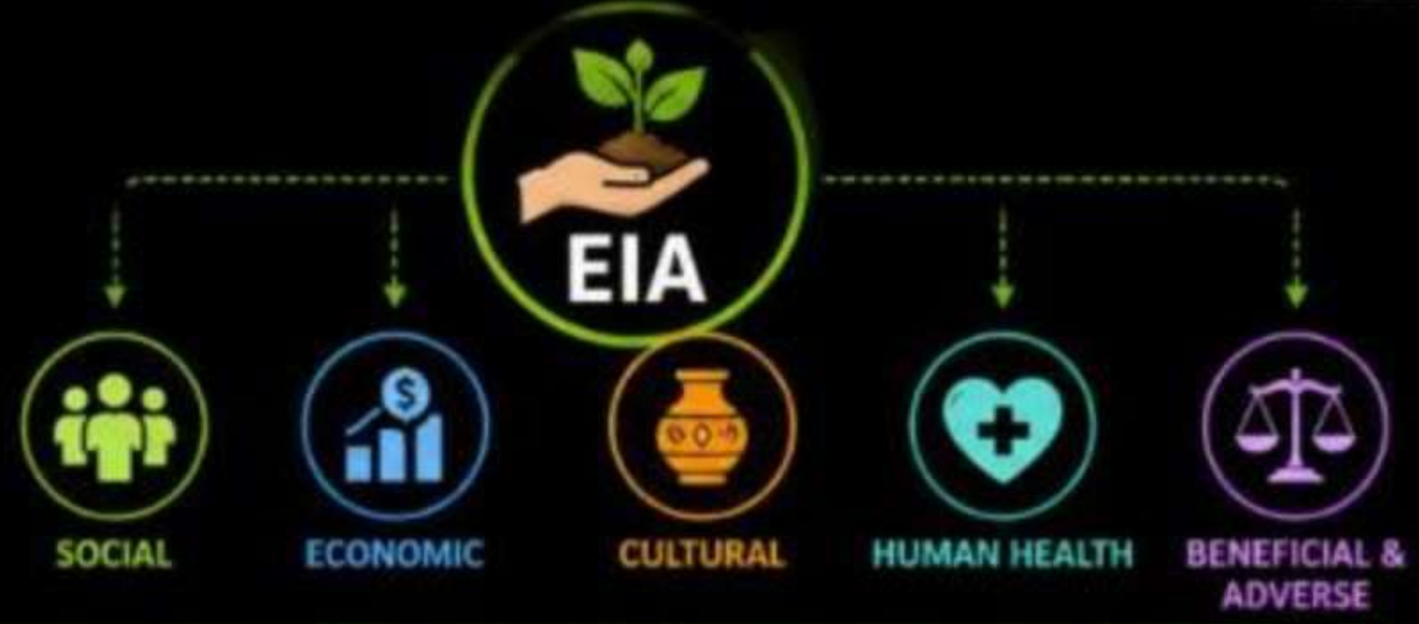
★ Together, let's build a sustainable, inclusive, and prosperous world for present and future generations. ★

ENVIRONMENTAL IMPACT ASSESSMENT (EIA)

(AKTU-2023-24,2017-18)

→ It is an **important management tool** for ensuring optimal use of natural resources for **sustainable development**.

→ It is a **process of evaluating** the environmental impacts of a proposed project or development taking inter-related socio-economic, cultural, and human-health impacts, **both beneficial and adverse**.



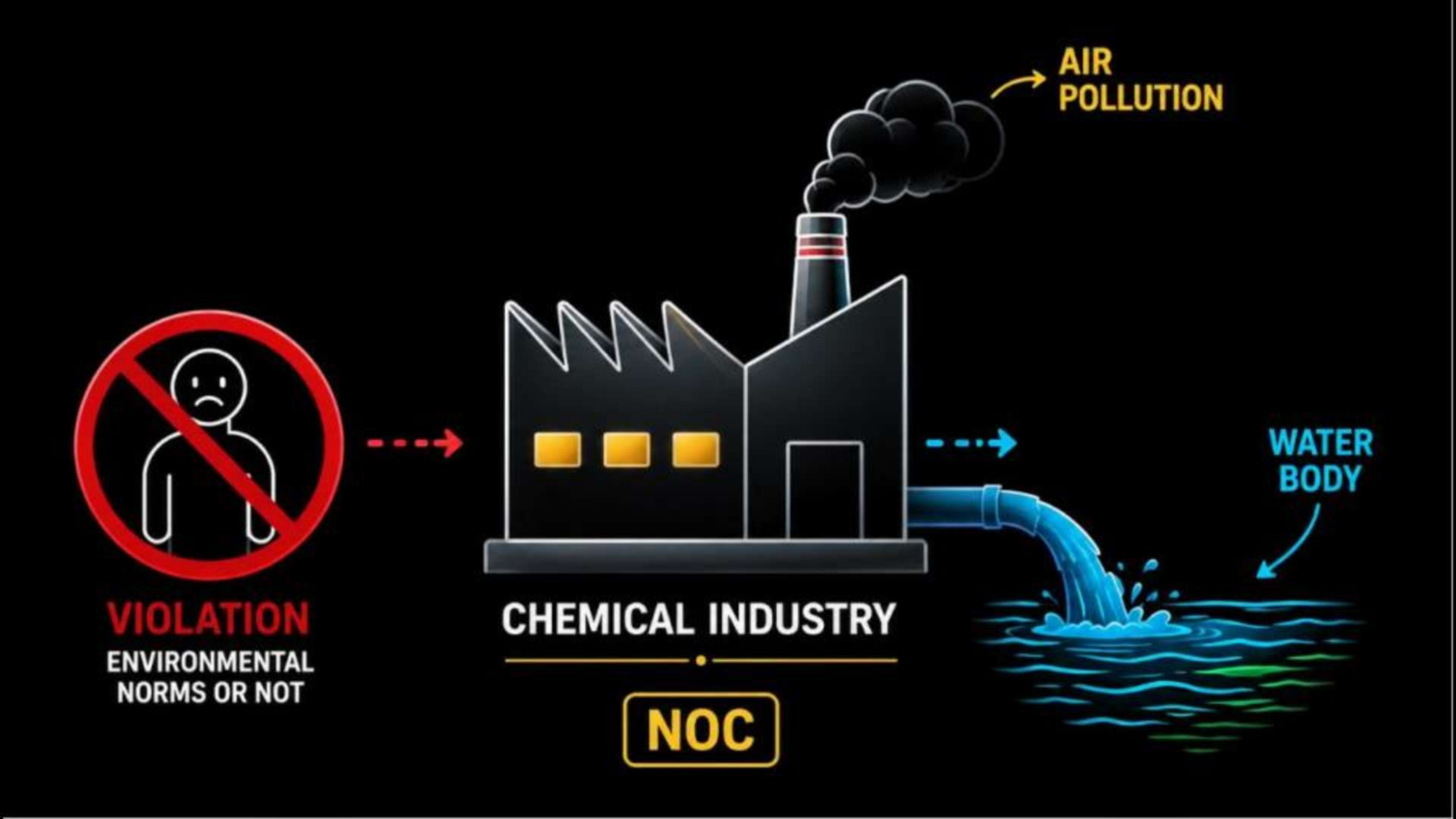
UNEP

United Nations Environment Program (UNEP) defines Environmental Impact Assessment (EIA) as a tool used to identify the environmental, social and economic impacts of a project prior to decision-making.

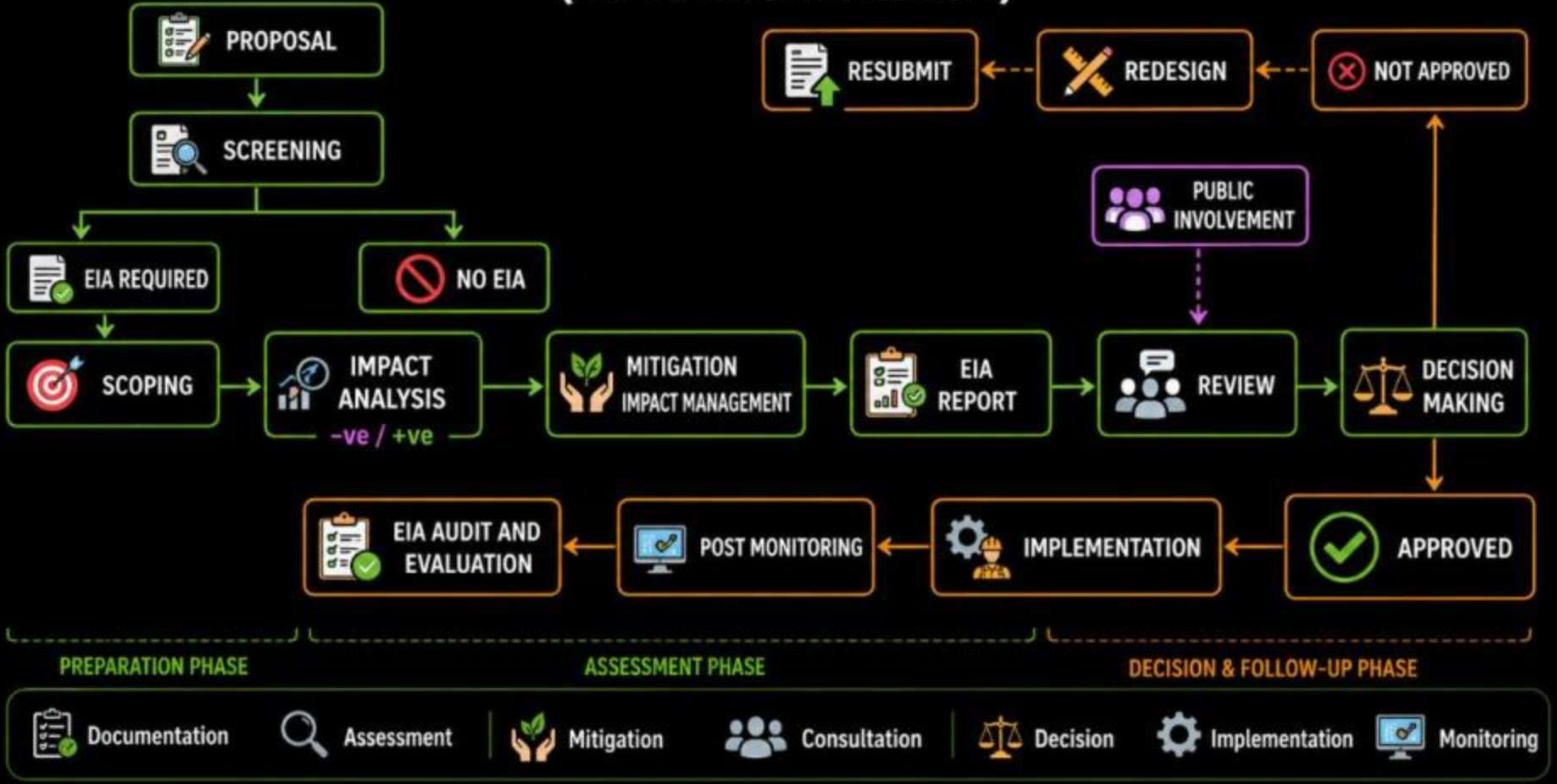
→ It aims to predict environmental impacts at an **early stage** in project planning and design, find ways and means to **reduce adverse impacts**, shape projects to suit the **local environment** and present the **predictions** and **options** to decision-makers.

EIA PROCESS – A PATH TO SUSTAINABLE DECISIONS

1. **SCREENING** — Identify if EIA is required
2. **SCOPING** — Identify key issues and impacts
3. **IMPACT ASSESSMENT** — Analyze potential impacts (beneficial & adverse)
4. **MITIGATION MEASURES** — Propose ways to avoid, minimize or compensate
5. **PUBLIC CONSULTATION** — Engage stakeholders and communities
6. **EIA REPORT** — Present findings and recommendations
7. **DECISION MAKING** — Informed and sustainable decision


AIR
POLLUTION
WATER
BODY
VIOLATION
ENVIRONMENTAL
NORMS OR NOT
CHEMICAL INDUSTRY
NOC

ENVIRONMENTAL IMPACT ASSESSMENT – FLOW CHART
(पर्यावरणीय प्रभाव आकलन)



IMPORTANCE OF EIA

01 EIA links environment with development for environmentally safe and sustainable development.

02 EIA provides a cost effective method to eliminate or minimize the adverse impact of developmental projects.

03 EIA enables the decision makers to analyze the effect of developmental activities on the environment well before the developmental project is implemented.

04 EIA encourages the adaptation of mitigation strategies in the developmental plan.

05 EIA makes sure that the developmental plan is environmentally sound and within the limits of the capacity of assimilation and regeneration of the ecosystem.

EIA ensures informed decisions, protects the environment, saves costs, and promotes sustainable development.

EFFECTS OF HUMAN ACTIVITIES ON ENVIRONMENT

FOOD (FOOD SECURITY)

- By the time the food you eat, much of the environmental impact has already occurred.
- Before food production even starts, natural habitats and ecosystems are destroyed to clear land that will be used for agriculture.

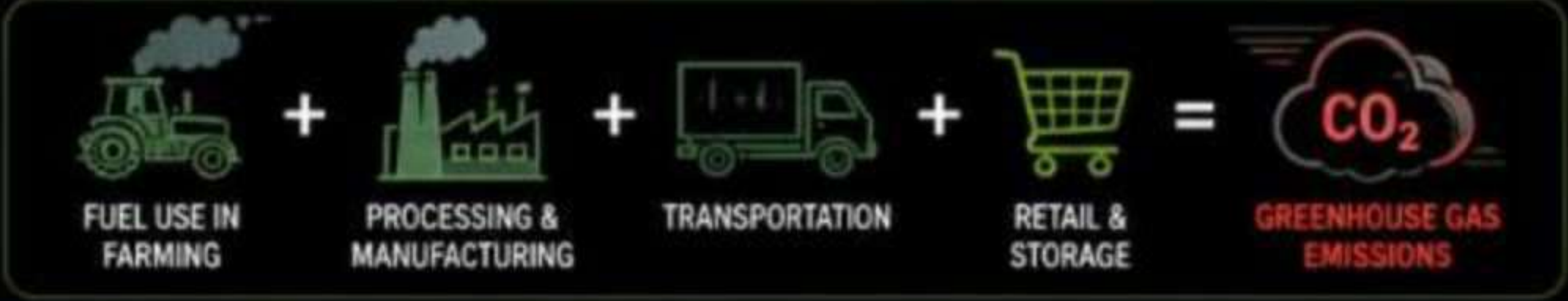
ENVIRONMENTAL IMPACTS OF FOOD PRODUCTION

1 WATER USE & WATER POLLUTION

- Growing food takes a lot of water. About 70% of all water use goes towards agricultural efforts. When runoff of agricultural pollutants occurs, groundwater get contaminated with things like nitrogen and phosphorus- commonly used in modern farming practices.

2 GREENHOUSE GAS EMISSIONS

- Greenhouse gas emissions, such as Carbon dioxide, are created when fossil fuels are used during several aspects of the food cycle, including food production and distribution.



Sustainable food systems, efficient resource use, and responsible choices can reduce environmental impact and ensure FOOD SECURITY for a better future.

CONTAMINANTS & POLLUTANTS

3 ENVIRONMENTAL CONTAMINANTS & POLLUTANTS

→ The growing, producing, and transporting of food can generate various contaminants that can have **adverse effects on the health of humans and the ecosystem**.

→ These contaminants include **ammonia** and the emission of different **nitrogen compounds** that disrupt the soil, animal and plant life.

DEPLETION OF NATURAL RESOURCES

4 DEPLETION OF NATURAL RESOURCES

→ Food production takes up a significant portion of the world's **natural resources**.

DEGRADATION OF LAND

5 DEGRADATION OF LAND

→ Harvesting the crop represents a significant amount of **nutrients**, **water**, and **energy** being taken from the land.

→ This leaves the **land barren**, and unfriendly for the growth and development of new organisms and ecosystems.

Sustainable food systems and **responsible resource use** are essential for **a healthy planet and a secure future.**

MAJOR WORLD FOOD PROBLEMS - There are two kinds of food insufficiency undernourishment and malnourishment.

1. UNDERNOURISHMENT (कुपोषण)

→ Undernourishment is the **lack of sufficient calories** in available food, so that one has little or no ability to move or work.

→ The Food and Agricultural Organization (FAO) estimates that the average minimum daily calorie intake is about **2500 calories per day.**

→ People who receive **less than 90%** of the minimum dietary intake on a long-term basis are considered **undernourished.**

→ Undernourished persons have **less energy** for doing any kind of work.

→ They are **susceptible to diseases**, their body becomes weak and they frequently fall sick.

TWO KINDS OF FOOD INSUFFICIENCY

DID YOU KNOW?

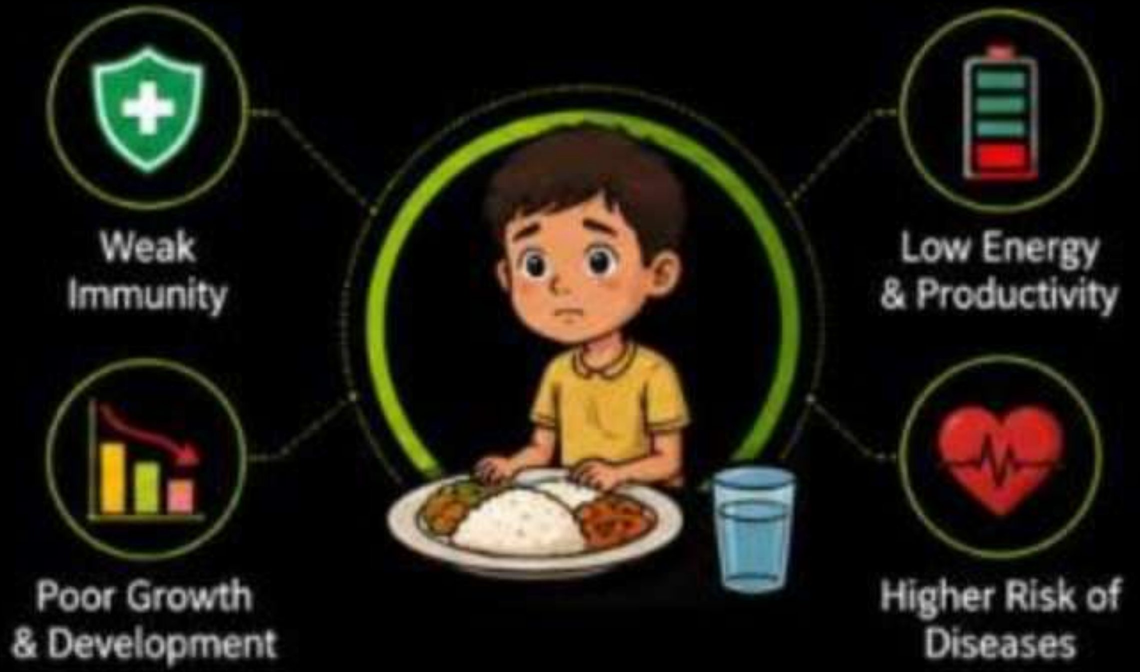
- ~735 million people worldwide are undernourished (FAO, 2023).
- Undernourishment affects **productivity**, **education**, and overall **quality of life**.
- Ensuring **food security** and **proper nutrition** is essential for a healthy and strong society.

2. MALNOURISHMENT (कुपोषण)

→ It means lack of essential nutrients (like proteins, vitamins, fats, minerals etc) in the diet.

→ It is possible to have excess food and still suffer from malnourishment due to nutritional imbalance caused by a lack of specific dietary components or an inability to absorb or utilize essential nutrients.

→ Some of the major problems of malnourishment are -



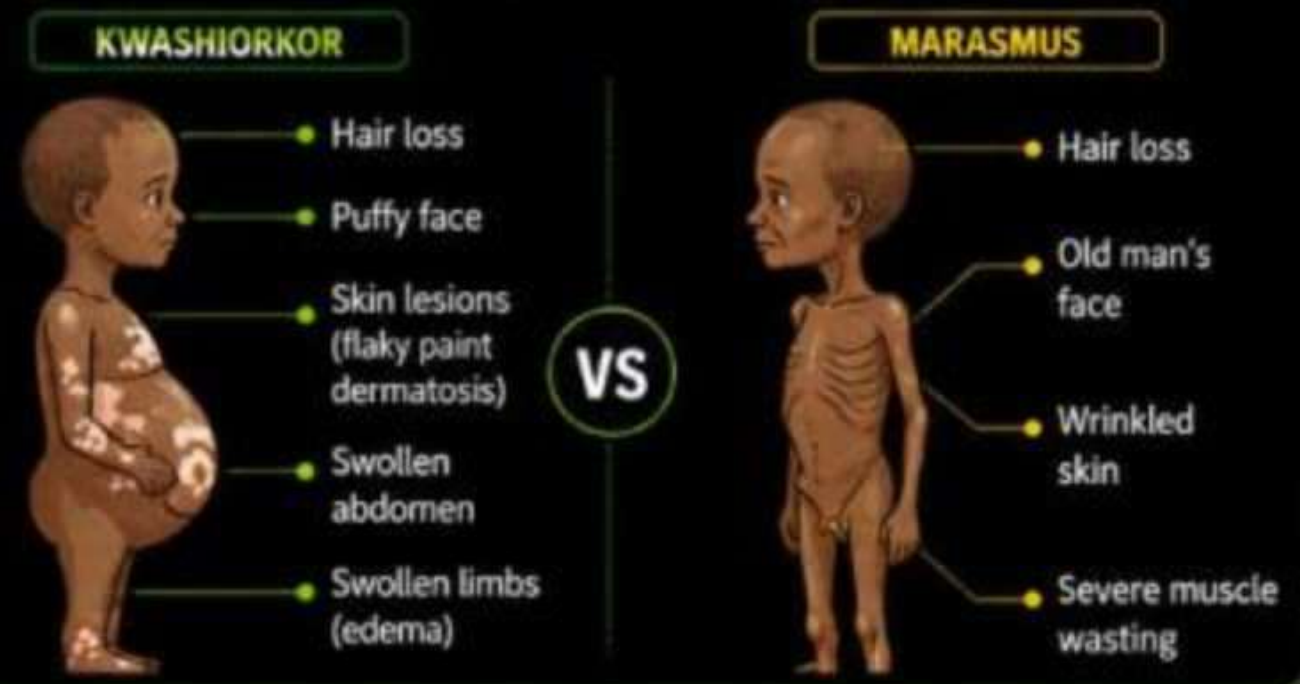
MARASMUS –

→ Marasmus is a kind of malnutrition that occurs due to **deficiency of proteins, carbohydrates and fats** in the diet.

→ In marasmus the child consumes **little or no food** leading to minimal calorie intake and also a total lack of nutrients.

→ Symptoms are **Weight loss, dehydration, slowing of growth, shrinkage of stomach.**

MARASMUS vs KWASHIORKOR



Good nutrition is the foundation of a healthy life.
Balanced diet, diverse foods, and proper care can **end malnutrition** and build a stronger future.

EFFECTS OF HUMAN ACTIVITIES ON ENVIRONMENT

AGRICULTURE

EFFECTS OF AGRICULTURE ACTIVITIES ON ENVIRONMENT

- Agriculture is the world's **oldest and largest industry** and more than half of all the people in the world still live on farms.
- Agriculture has both **primary and secondary environmental effects**.
- A **primary effect** is an effect on the area where the agriculture takes place, i.e. **on-site effect**.

ON-SITE EFFECT

- A **secondary effect**, also called an **off-site** effect, is an effect on an environment away from the agricultural site.

OFF-SITE EFFECT

AGRICULTURE'S IMPACT ON THE ENVIRONMENT

- Deforestation & Habitat Loss
- Soil Degradation & Erosion
- Water Pollution & Overuse
- Greenhouse Gas Emissions
- Loss of Biodiversity

ON-SITE vs OFF-SITE EFFECTS

ON-SITE EFFECT

Impacts occur on the area where **farming activities** take place.

- Soil nutrients depletion
- Pesticide & fertilizer residues
- Soil erosion & compaction
- Loss of soil organic matter

OFF-SITE EFFECT

Impacts occur outside the farming area, affecting the **surrounding environment**.

- Water contamination
- Air pollution & drift
- Eutrophication of water bodies
- Impact on wildlife & humans

Sustainable agriculture practices, responsible resource use and **environmental conservation** can minimize negative impacts and ensure **food security** for future generations.

SUSTAINABLE PRACTICES → RESPONSIBLE RESOURCE USE → ENVIRONMENTAL CONSERVATION → FOOD SECURITY

PESTICIDES

→ A pesticide is a chemical substance used to kill or control pests, such as insects, weeds, fungi, and rodents.

→ Pesticides help protect crops from damage and increase agricultural productivity.

BENEFITS OF PESTICIDES

- Protect crops from pests
- Increase agricultural productivity
- Improve crop quality and yield
- Support food security and farmers' income

PESTICIDES-RELATED PROBLEMS

1 HARM TO OTHER ANIMALS

→ Pesticides intended to kill pests can also harm other animals like birds, and fish.

→ This reduces the population of beneficial insects (like pollinators) and other wildlife.

Disturbs ecological balance and reduces biodiversity.

2 HEALTH RISKS FOR PEOPLE

→ People can get exposed to pesticides by breathing them in, touching them, or eating food with pesticide residues.

→ This exposure can cause immediate health issues like poisoning or long-term problems like cancer and neurological disorders.

POSSIBLE HEALTH EFFECTS

- Poisoning (Acute effects)
- Cancer (Long-term)
- Neurological Disorders
- Digestive Problems
- Skin Irritation

Use pesticides responsibly. Follow recommended doses, ensure proper storage and disposal, and promote integrated pest management for a safe and sustainable future.

EFFECTS OF HUMAN ACTIVITIES ON THE ENVIRONMENT:

TRANSPORTATION

Transportation is a **major human activity** that significantly impacts the environment.

AIR POLLUTION

- **Cause:** Emissions from cars, trucks, airplanes, and ships.
- **Effect:** Release of harmful pollutants like carbon monoxide, nitrogen oxides, and particulate matter, which contribute to air quality degradation and health problems like asthma and lung disease.

POLLUTANTS RELEASED

- CO – CARBON MONOXIDE
- NO_x – NITROGEN OXIDES
- PARTICULATE MATTER

GREENHOUSE GAS EMISSIONS

- **Cause:** Burning of fossil fuels for transportation.
- **Effect:** Increased levels of carbon dioxide (CO_2) and other greenhouse gases in the atmosphere, contributing to global warming and climate change.

CONTRIBUTES TO

- GLOBAL WARMING & CLIMATE CHANGE
- RISING TEMPERATURES & EXTREME WEATHER

NOISE POLLUTION

- **Cause:** Traffic, aircraft, and other transportation-related activities.
- **Effect:** Disturbance to wildlife and humans, leading to stress, hearing loss, and reduced quality of life.

HABITAT DESTRUCTION

- **Cause:** Construction of roads, highways, airports, and railways.
- **Effect:** Loss of natural habitats for animals and plants, leading to reduced biodiversity.

WATER POLLUTION

- **Cause:** Runoff from roads and highways, and oil spills from ships.
- **Effect:** Contamination of water bodies, harming aquatic life and affecting water quality.

LAND USE

- **Cause:** Expansion of infrastructure for transportation.
- **Effect:** Reduction of green spaces and agricultural land, impacting ecosystems and local communities.

Sustainable transportation, cleaner fuels, noise control, responsible infrastructure planning, and protection of natural habitats can **reduce environmental impacts** and create a **healthier and sustainable future.**

- CLEANER FUELS
- PUBLIC TRANSPORT
- NOISE CONTROL
- ECO-FRIENDLY INFRASTRUCTURE
- PROTECT OUR PLANET

EFFECTS O‑HUMAN ACTIVITIES ON THE ENVIRONMENT: HOUSING

Housing development has a significant impact on the environment.

1. DEFORESTATION AND HABITAT DESTRUCTION

Cause: Clearing land for building omes and communities.

Effect: Loss of forests and natural habitats, reducing biodiversity and disrupting ecosystems.

FOREST → LAND CLEARING → HOUSING DEVELOPMENT → LOSS OF BIODIVERSITY

2. RESOURCE CONSUMPTION

Cause: Use of materials like wood, concrete, and metals in construction.

Effect: Depletion of natural resources and increased environmental degradation from mining and logging.

WOOD + CONCRETE + METALS = CONSTRUCTION → RESOURCE DEPLETION

3. ENERGY USE

Cause: Heating, cooling, lighting, and powering homes.

Effect: Increased demand for energy, often generated from fossil fuels, leading to higher greenhouse gas emissions.

ENERGY USE (Electricity & Fuel) → FOSSIL FUEL POWER PLANTS → INCREASED CO_2 EMISSIONS → CLIMATE CHANGE & GLOBAL WARMING

Sustainable housing, efficient resource use, and clean energy can reduce environmental impacts and build a better future.

GREEN BUILDING | ENERGY EFFICIENCY | RENEWABLE ENERGY | WATER CONSERVATION | REDUCE WASTE | PROTECT NATURE

OTHER ENVIRONMENTAL IMPACTS RELATED TO HOUSING

4. WATER CONSUMPTION

Cause: Daily household use, landscaping, and construction processes.

Effect: Increased pressure on local water supplies and potential for water shortages.

HOUSEHOLD USE & LANDSCAPING → PRESSURE ON WATER SUPPLIES → WATER SHORTAGE

5. WASTE GENERATION

Cause: Construction, household waste, and disposal of old materials.

Effect: Increased landfill use, pollution from improper disposal, and greater need for waste management systems.

WASTE GENERATION → IMPROPER DISPOSAL & COLLECTION → LANDFILL USE & POLLUTION

6. POLLUTION

Cause: Use of chemicals in construction and household products.

Effect: Release of pollutants into air, water, and soil, affecting health and ecosystems.

CHEMICALS & CONSTRUCTION ACTIVITIES → POLLUTANTS RELEASED INTO AIR, WATER & SOIL → HEALTH ISSUES & ECOSYSTEM DAMAGE

Responsible planning, conservation of resources, proper waste management, and pollution control can create sustainable and healthy communities.

SUSTAINABLE PLANNING | RESOURCE CONSERVATION | PROPER WASTE MANAGEMENT | POLLUTION CONTROL | HEALTHY COMMUNITIES


AKTU
ALL BRANCHES
Gateway
AARMBH BATCH
2.0
Sem-I + Sem-II
COMBO of ALL Subject
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Thank You

For a Better Future, Let's Choose **Sustainability.**